

Differentially Private Preference Data Synthesis for Large Language Model Alignment

Anonymous Authors¹

Abstract

Preference alignment is a central post-training paradigm for large language models (LLMs), spanning settings in which model behavior is adapted to human judgments, task requirements, and deployment norms. Yet alignment on real preference data can expose sensitive information because such datasets may contain user prompts, candidate responses, and annotator choices that reflect private attributes or beliefs. We propose **DPPrefSyn**, an algorithm for generating differentially private (DP) synthetic preference data that supports privacy-preserving preference alignment without changing the downstream alignment method. DPPrefSyn provides a principled framework grounded in the Bradley–Terry preference model and the geometric structure of pairwise preference data. It learns a private preference representation with formal DP guarantees, then combines the learned preference model with public prompts to synthesize preference pairs. The method uses shared linear structure across per-cluster reward models to represent heterogeneous preference patterns and applies DP Principal Component Analysis (DP-PCA) to improve accuracy in high-dimensional embedding spaces. Across question answering and summarization benchmarks, DPPrefSyn attains competitive alignment performance under strong DP guarantees. These results support DP synthetic preference data as a broadly reusable alternative for privacy-preserving alignment in scenarios where preference data are sensitive but public prompts or task inputs are available. To the best of our knowledge, this is the first work to generate DP synthetic preference data for LLM alignment. Our code will be released upon acceptance.

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

1. Introduction

Preference alignment methods, including RLHF (Stiennon et al., 2020) and DPO (Rafailov et al., 2024), have become a general post-training tool for adapting LLMs to human expectations across domains. These methods use pairwise preference data in which annotators compare candidate responses and identify the response that better satisfies the prompt. The resulting comparisons guide fine-tuning by encouraging the model to assign higher preference to selected responses than to rejected ones (Ziegler et al., 2019; Ouyang et al., 2022; Bai et al., 2022). Because this paradigm can be applied wherever human judgments over model outputs are available, it has been adopted in settings ranging from chat assistants (Achiam et al., 2023) to mathematical reasoning tools (Shao et al., 2024) and code generation systems (Shen et al., 2023).

At the same time, preference alignment creates privacy risks across the full preference record, not only in labels or model updates. Real preference datasets often include user prompts and human feedback; prompts can reveal information about health, identity, or other sensitive topics, while feedback can encode private beliefs, preferences, or behavioral patterns (Li et al., 2023; Yu et al., 2024). Prior work (Chowdhury et al., 2024; Yu et al., 2024; Wu et al., 2023a) has used differential privacy (DP) (Dwork et al., 2006b) to reduce these risks and has produced promising results. However, many existing approaches protect only one component of the record, such as prompts (Yu et al., 2024) or preference labels (Chowdhury et al., 2024; Zhang et al., 2025), leaving other components available for inference. Other methods are tied to a particular alignment algorithm, such as RLHF (Wu et al., 2023a), rather than producing reusable data that can support newer objectives such as DPO. These limitations become especially important when private preference annotations are scarce, costly to collect, or difficult to share across models and applications. These gaps motivate the central question of this work:

Is it possible to achieve superior preference alignment while ensuring thorough privacy protection and maintaining compatibility with emerging alignment methodologies?

We answer this question affirmatively by developing a

reusable DP data-synthesis approach for preference alignment. Our main contributions are summarized as follows:

- We introduce DPPrefSyn (Figure 1), an algorithm that generates DP synthetic preference data for LLM alignment. The framework is grounded in the Bradley–Terry preference model and in the geometry of pairwise human preference data, making it applicable to preference datasets represented as prompt, preferred response, and dispreferred response triples. It features the following design elements: 1) *Capturing diverse human preferences via shared linear structure of rewards and clustering*. The Bradley–Terry preference model, together with a linear reward structure, allows us to cluster preference samples by embedding differences between concatenated prompt–preferred and prompt–dispreferred pairs, so that each cluster can be represented by a shared reward function. 2) *Improving learning efficiency through DP-PCA*. To make clustering and reward estimation practical in high-dimensional embedding spaces, we use DP-PCA to project embeddings into a compact subspace while preserving preference-relevant variation under DP. 3) *Saving privacy budgets by using public prompts*. We use public prompts to generate candidate responses and apply private reward models to select preference pairs, allocating the privacy budget to private preference modeling rather than to prompt generation.
- We provide a privacy analysis showing that DPPrefSyn satisfies (ϵ, δ) -DP for the private preference dataset. The analysis assigns privacy budget ϵ_0 to DP-PCA (Amin et al., 2019) for dimensionality reduction and ϵ_1 to DP-KMeans (Su et al., 2016) for clustering, with the remaining budget used for per-cluster reward training via DP-SGD. We use the Privacy Random Variable (PRV) accountant (Gopi et al., 2021) to compose the DP-SGD cost tightly. By DP post-processing, the synthetic dataset can then be reused across alignment algorithms and LLM families without additional privacy cost.
- We evaluate DPPrefSyn on standard benchmarks covering assistant-style question answering from OpenAssistant (Köpf et al., 2023), helpfulness and harmlessness comparisons from Anthropic-HH (Bai et al., 2022), and TL;DR summarization (Stiennon et al., 2020). These experiments examine whether the same synthesis pipeline supports distinct tasks, datasets, privacy budgets, and downstream model classes. Under privacy budgets $\epsilon = 0.5, 1, 2, 4, 8$, DPPrefSyn consistently outperforms DP fine-tuning on real private data, and in many cases also exceeds non-private fine-tuning on real data. Additional studies show that DPPrefSyn remains effective across alignment algorithms, downstream LLMs, public data scales, embedding choices, and empirical membership inference attacks.

2. Related Work

Differentially Private Alignment of LLMs. Wu et al. (2023a) introduce a DP framework for LLM alignment with reinforcement learning by adapting PPO to the DP setting. This demonstrates that DP can be incorporated into alignment, but the approach is *limited to PPO* even as preference alignment methods continue to diversify. Chowdhury et al. (2024) study reward estimation from preference-based feedback through label-DP (Ghazi et al., 2021), and Zhang et al. (2025) propose AUP-RLHF, a user-level label-DP framework for RLHF. These methods protect annotator labels, but they *do not address the sensitivity of prompts or responses themselves*. They also remain *constrained by the limited private user preference data*, which can limit alignment quality when human annotations are expensive to obtain. Yu et al. (2024) protect sensitive user *instructions* by generating DP synthetic *instructions* to replace real ones during data annotation and model fine-tuning.

Beyond preference alignment, related privacy techniques address other LLM adaptation settings, including private fine-tuning on general labeled data rather than pairwise comparisons (Yu et al., 2021; Li et al., 2021; Chen et al., 2024; Tang et al., 2024; Zhang et al., 2023; He et al., 2022) and privacy protection for in-context prompts in in-context learning (Duan et al., 2023; Tang et al., 2023; Gao et al., 2024; Hong et al., 2023; Wu et al., 2023b).

Differentially Private Synthetic Text Generation. Our work belongs to the broader area of DP synthetic text generation, where the goal is to release or reuse text-like data while limiting disclosure of the private source corpus. One line of work fine-tunes a language model on private data under DP and then samples synthetic text from the private fine-tuned model (Yue et al., 2023; Mattern et al., 2022; Mireshghallah et al., 2023; Carranza et al., 2023; Yu et al., 2024; Wang et al., 2024; Ochs & Habernal, 2024; Tan et al., 2025; Carranza et al., 2024). For example, Yu et al. (2024) privately fine-tune the LLaMA model on private instruction data and use the resulting model to generate synthetic text for supervised instruction tuning.

Because DP fine-tuning can be expensive or unavailable for non-public language models, another line of work generates DP synthetic data by prompting LLM APIs (Lin et al., 2023; Xie et al., 2024; Wu et al., 2024; Hou et al., 2024). These methods typically use API access to propose synthetic samples and then select or refine outputs according to privacy-preserving similarity to private data. For instance, Xie et al. (2024) propose Aug-PE, which iteratively refines LLM-generated samples using embedding-space similarity to private data. *Our work follows this API-compatible synthesis perspective but targets pairwise preference data for LLM alignment rather than ordinary text or instruction data.*

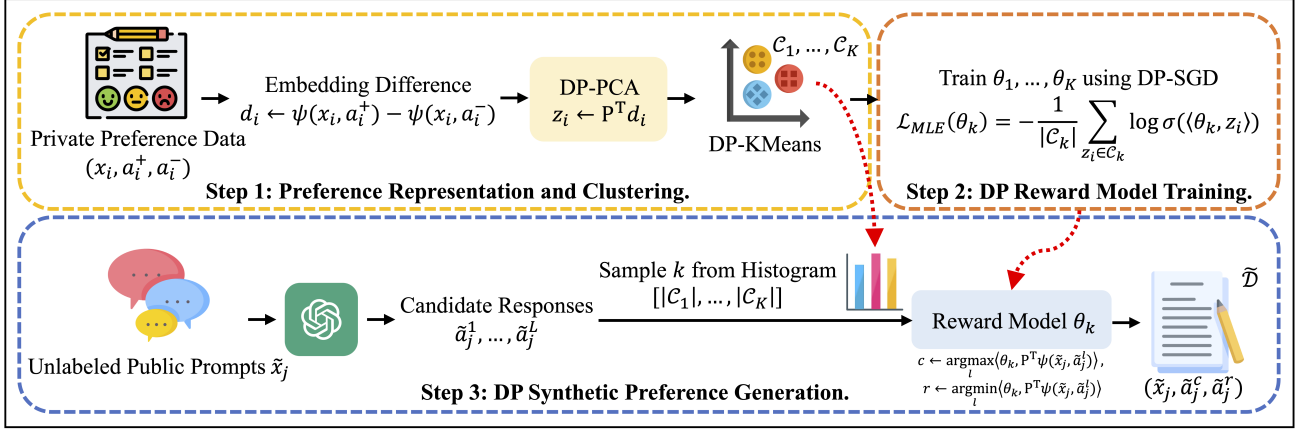


Figure 1. Overview of DPPrefSyn. DPPrefSyn generates DP synthetic preference data through 3 reusable stages: 1) representing preference samples as embedding differences and clustering them via DP-PCA and DP-KMeans, 2) training DP reward models on each cluster using DP-SGD, and 3) generating synthetic preference samples from public prompts guided by the DP-protected distribution of reward functions.

Diversity in Human Preferences. Preference data often reflects heterogeneous human judgments rather than a single universal ordering (Denton et al., 2021; Aroyo et al., 2023a;b; Chakraborty et al., 2024). Such diversity can arise from sociodemographic differences (e.g., race, gender, age), personal beliefs and biases, domain expertise, and ambiguity in natural language (Sandri et al., 2023; Vogels, 2021). Motivated by this broader view, recent work has studied personalized preference alignment for different users or groups (Lee et al., 2024b; Poddar et al., 2024; Singh et al., 2025). For example, Singh et al. (2025) propose a meta-learning framework in which an LLM adapts to individual users from a small number of preference examples. Our approach generates DP synthetic preference data that *captures the underlying diversity of human feedback*, improving alignment relative to variants that collapse this diversity (as shown in Section 5.3).

3. Problem Definition

Preference Alignment of LLMs. We study preference alignment for an LLM using pairwise human feedback, a data format that appears across many alignment tasks. The preference dataset is $\mathcal{D} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$. Each sample contains a prompt x_i and two responses a_i^+, a_i^- , where annotators prefer a_i^+ to a_i^- .

Following prior work on reinforcement learning from human feedback (RLHF) and direct preference optimization (DPO) (Ouyang et al., 2022; Zhu et al., 2023; Rafailov et al., 2024; Liu et al., 2023), we use the Bradley–Terry model (Bradley & Terry, 1952) as a common probabilistic descrip-

tion of pairwise preferences:

$$\begin{aligned} \mathbb{P}[a_i^+ \succ a_i^- \mid x_i] &= \frac{\exp(r^*(x_i, a_i^+))}{\exp(r^*(x_i, a_i^+)) + \exp(r^*(x_i, a_i^-))} \\ &= \sigma(r^*(x_i, a_i^+) - r^*(x_i, a_i^-)), \end{aligned}$$

where $a_i^+ \succ a_i^-$ means a_i^+ is preferred to a_i^- , $r^*(x, a)$ is a latent reward function, and $\sigma(z) = 1/(1 + e^{-z})$ is the sigmoid function.

This same formulation supports multiple alignment objectives: RLHF explicitly learns a reward model from preference data, while DPO induces an implicit reward through the log-ratio between the learned policy and a reference policy.

Differentially Private Preference Data Synthesis. Our goal is to protect the private preference dataset $\mathcal{D}_{\text{priv}} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$ against adversaries who might infer information about individual prompts, responses, or preference labels from a fine-tuned LLM or released training artifact. We therefore generate a DP synthetic preference dataset $\tilde{\mathcal{D}} = \{(\tilde{x}_j, \tilde{a}_j^+, \tilde{a}_j^-)\}_{j=1}^m$ that retains preference information useful for alignment while satisfying formal privacy guarantees for the source data. By the post-processing property of DP, $\tilde{\mathcal{D}}$ can be reused across preference alignment methods and LLMs without additional privacy cost.

We use the standard definition of (ϵ, δ) -differential privacy.

Definition 3.1 ((ϵ, δ) -Differential Privacy (Dwork et al., 2006a)). A randomized algorithm $\mathcal{A}$ is (ϵ, δ) -differentially private if for any two neighboring inputs $\mathcal{D}$ and $\mathcal{D}'$ that differ by a single entry and any set $\mathcal{S}$ of possible outputs: $\mathbb{P}[\mathcal{A}(\mathcal{D}) \in \mathcal{S}] \leq e^\epsilon \mathbb{P}[\mathcal{A}(\mathcal{D}') \in \mathcal{S}] + \delta$.

Diverse User Preferences and Linear Reward Structure. Private preference datasets typically aggregate judg-

ments from non-homogeneous annotators or user groups (Chakraborty et al., 2024; Bakker et al., 2022; Kovač et al., 2023; Jang et al., 2023; Rame et al., 2023; Ji et al., 2023). Annotators may emphasize factual accuracy, politeness, creativity, clarity, or other task-dependent qualities, creating diverse and sometimes conflicting signals. A synthetic dataset $\tilde{\mathcal{D}}$ is therefore most useful when it preserves this diversity rather than representing the private data as a single averaged preference.

To model this setting while keeping private learning tractable, we assume a linear structure for the latent reward function (Saha et al., 2023; Kong & Yang, 2022; Zhu et al., 2023). Specifically, we adopt a linear reward model

$$r_\theta(x, a) = \langle \theta, \phi(x, a) \rangle,$$

where $\phi(x, a) \in \mathbb{R}^d$ is the *shared* feature vector and $\theta \in \mathbb{R}^d$ is a *user-dependent* parameter.

The linear structure balances *expressiveness* and *tractability* for heterogeneous preference modeling. Users or groups can share semantic features in an embedding space (i.e., $\phi(x, a)$) while differing in how they weight those features (i.e., θ). This parameterization is broad enough to represent multiple preference clusters yet simple enough to learn under limited data and strong privacy constraints. As shown in Section B.6, the linear reward structure captures preference signals effectively and performs competitively with a fully fine-tuned GPT-2 reward model despite its simplicity.

4. Proposed Method

4.1. Key Components of the Algorithm Design

Before presenting the full algorithm, we identify the design components that make DP preference synthesis applicable across preference datasets with heterogeneous judgments, high-dimensional text embeddings, and limited private annotations.

Leveraging Linear Reward Structure for Preference Clustering. The linear reward model indicates that

$$\mathbb{P}[a_i^+ \succ a_i^- \mid x_i] = \sigma(\langle \theta, \phi(x_i, a_i^+) - \phi(x_i, a_i^-) \rangle).$$

For a tuple (x_i, a_i^+, a_i^-) , $a_i^+ \succ a_i^-$ implies that $\langle \theta, \phi(x_i, a_i^+) - \phi(x_i, a_i^-) \rangle \gg 0$, so the preference vector θ aligns with the feature difference $\phi(x_i, a_i^+) - \phi(x_i, a_i^-)$. Preference samples produced by annotators or groups with similar criteria should therefore induce difference vectors that point in similar directions in the embedding space. We use this geometry to cluster preference samples by $\phi(x_i, a_i^+) - \phi(x_i, a_i^-)$, allowing preferences within each cluster to be approximated by a shared reward function with a common preference vector θ . In Section B.9, we show that the resulting clusters are meaningful and interpretable as preference styles.

Adopting DP-PCA to Obtain Effective Features. A natural feature vector $\phi(x, a)$ is the embedding of the corresponding (prompt, response) pair, since embeddings encode semantic and stylistic information that can drive preference judgments. Such embeddings are often high-dimensional (e.g., 1024 dimensions), reflecting the richness of text but also increasing the sample requirements for clustering and reward learning. Because each cluster may contain only a subset of the private data, directly learning in the original embedding space can be inefficient. We therefore apply differentially private PCA (DP-PCA) (Amin et al., 2019; Liu et al., 2022) to obtain compact representations for both clustering and reward model training. This step improves sample efficiency and clustering stability while retaining preference-relevant signals for downstream alignment.

Utilizing Public Prompts to Save DP Budget. Jointly synthesizing prompts, preferred responses, and less preferred responses would spend privacy budget on every component of the record and can reduce utility. We instead use public prompts when available, which lets the privacy budget focus on modeling preferences over responses. For each public prompt, an LLM generates multiple candidate responses, and a DP-protected distribution of reward functions learned from private preferences selects synthetic preference pairs. Although public prompts may come from a different distribution than the private prompts, our experiments show that this design can still preserve useful preference patterns and support downstream alignment tasks.

4.2. DPPrefSyn Algorithm

We now present DPPrefSyn (Algorithm 1), which transforms a private preference dataset $\mathcal{D}_{\text{priv}}$ into a DP synthetic preference dataset that can be consumed by different alignment pipelines. The algorithm has 3 main steps:

Step 1: Preference Representation and Clustering. We encode each prompt-response pair in $\mathcal{D}_{\text{priv}}$ with a public sentence embedding model ψ , producing a representation that is independent of any specific downstream LLM. For each sample (x_i, a_i^+, a_i^-) , we concatenate the prompt with each response to form $[x_i; a_i^+]$ and $[x_i; a_i^-]$, compute their embeddings, and define the preference difference $d_i \leftarrow \psi(x_i, a_i^+) - \psi(x_i, a_i^-)$ (Line 3).

We then reduce dimensionality privately by applying DP-PCA (Amin et al., 2019) to $\{d_i\}_{i=1}^n$, obtaining a projection matrix $\mathbf{P} \in \mathbb{R}^{d \times p}$ under ϵ_0 -DP (Line 5). At a high level, DP-PCA (Amin et al., 2019) approximates the eigendecomposition of the data covariance matrix by estimating eigenvalues and eigenvectors with privacy-preserving noise. Each d_i is projected to $z_i \leftarrow \mathbf{P}^\top d_i$ (Line 6). We then apply DP-KMeans (Su et al., 2016) to $\{z_i\}_{i=1}^n$ to group samples with similar preference patterns under ϵ_1 -DP, forming K clusters $\mathcal{C}_1, \dots, \mathcal{C}_K$ in Line 7. DP-KMeans (Su et al., 2016)

can be viewed as a DP version of Lloyd’s algorithm.

Step 2: DP Reward Model Training. For each cluster C_k , we learn a shared preference vector $\theta_k \in \mathbb{R}^p$ that represents the preference pattern of that cluster by minimizing the negative log-likelihood (Line 9):

$$\mathcal{L}_{\text{MLE}}(\theta_k) = -\frac{1}{|C_k|} \sum_{z_i \in C_k} \log \sigma(\langle \theta_k, z_i \rangle), \quad (1)$$

where z_i is the DP-PCA projected embedding difference from Step 1. Each reward model is trained with DP-SGD, a DP variant of SGD (Abadi et al., 2016), using noise multiplier σ_2 .

Step 3: DP Synthetic Preference Generation. We compute a histogram $\mathbf{h} = [|C_1|, \dots, |C_K|]$ of cluster sizes (Line 11) and normalize it into a cluster sampling distribution $\mathbf{p}$, i.e., $\mathbf{p} \leftarrow \mathbf{h}/|\mathcal{D}_{\text{priv}}|$ (Line 12). This distribution transfers the mixture of private preference styles to synthetic data generation.

For each public prompt $\tilde{x}_j$, we prompt an LLM at a relatively high temperature to produce L diverse candidate responses $\tilde{a}_j^1, \dots, \tilde{a}_j^L$ (Line 14). We sample a cluster index $k \sim \mathbf{p}$ from the DP histogram (Line 15) and score the candidates with the reward model parameterized by θ_k . Specifically, we compute embeddings $\psi(\tilde{x}_j, \tilde{a}_j^l)$ for $l = 1, \dots, L$, project them with $\mathbf{P}$, and assign reward $\langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$. The highest- and lowest-scoring responses become the preferred and less preferred responses: $c \leftarrow \arg \max_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$, $r \leftarrow \arg \min_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ (Lines 16-17). If the reward gap is too small (e.g., < 0.5), the pair is discarded to maintain preference quality; otherwise, $(\tilde{x}_j, \tilde{a}_j^c, \tilde{a}_j^r)$ is added to $\tilde{\mathcal{D}}$. Repeating this process over public prompts yields the DP synthetic preference dataset $\tilde{\mathcal{D}}$.

Privacy Analysis of DPPrefSyn. DPPrefSyn spends privacy budget only when it accesses the private preference dataset. We allocate ε_0 to DP-PCA (Line 5) and ε_1 to DP-KMeans (Line 7); the remaining budget is used to train per-cluster reward models with DP-SGD and noise multiplier σ_2 (Line 9). For reward training, disjoint clusters allow parallel composition: changing one data point affects only one cluster, so the privacy cost is controlled by the smallest cluster used for training. We discard clusters with too few samples to maintain a known lower bound on cluster size. The Privacy Random Variable (PRV) accountant (Gopi et al., 2021) composes the DP-SGD steps so that reward training satisfies $(\varepsilon - \varepsilon_0 - \varepsilon_1, \delta)$ -DP. All remaining operations use public data or DP post-processing and add no privacy cost. We provide the full proof and privacy-accounting code in Section A.

Discussion on Prompt Distribution Shift. DPPrefSyn is designed to remain useful when public prompts and pri-

Algorithm 1 DPPrefSyn

```

1: Input: Private dataset  $\mathcal{D}_{\text{priv}} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$ , public prompts  $\{\tilde{x}_j\}_{j=1}^m$ , embedding model  $\psi$ , number of clusters  $K$ , an LLM  $\text{LLM}(\cdot)$ , DP parameters  $\varepsilon_0, \varepsilon_1, \sigma_2$ .
2: for each  $(x_i, a_i^+, a_i^-)$  in  $\mathcal{D}_{\text{priv}}$  do
3:    $d_i \leftarrow \psi(x_i, a_i^+) - \psi(x_i, a_i^-)$ , where  $d_i \in \mathbb{R}^d$ 
4: end for
5:  $\mathbf{P} \leftarrow \text{DP-PCA}(\{d_i\}_{i=1}^n, \varepsilon_0)$ , where  $\mathbf{P} \in \mathbb{R}^{d \times p}$ 
6:  $\{z_i\}_{i=1}^n \leftarrow \{\mathbf{P}^\top d_i\}_{i=1}^n$ 
7:  $C_1, \dots, C_K \leftarrow \text{DP-KMeans}(\{z_i\}_{i=1}^n, K, \varepsilon_1)$ 
8: for each cluster  $C_k$  do
9:   Train a linear reward model  $\theta_k$  on  $C_k$  using DP-SGD with noise multiplier  $\sigma_2$  (Equation (1))
10: end for
11:  $\mathbf{h} \leftarrow [|C_1|, \dots, |C_K|]$ 
12:  $\mathbf{p} \leftarrow \mathbf{h}/|\mathcal{D}_{\text{priv}}|$ 
13: for each public prompt  $\tilde{x}_j$  do
14:   Generate responses  $\tilde{a}_j^1, \dots, \tilde{a}_j^L \leftarrow \text{LLM}(\tilde{x}_j)$ 
15:   Sample cluster index  $k \sim \mathbf{p}$ 
16:    $c \leftarrow \arg \max_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ 
17:    $r \leftarrow \arg \min_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ 
18:   Add  $(\tilde{x}_j, \tilde{a}_j^c, \tilde{a}_j^r)$  to  $\tilde{\mathcal{D}}$ 
19: end for
20: return  $\tilde{\mathcal{D}}$ .
```

ate prompts are not identically distributed. The preference diversity is represented by θ_k , the parameter of the linear reward model, rather than solely by $\phi(x, a)$, the feature vector for a particular prompt-response pair. Intuitively, a user may ask many different prompts while maintaining a stable underlying preference style. Thus, if the distribution of θ_k matches the private data, synthetic preferences can still reflect private preference patterns on public prompts. DPPrefSyn implements this by sampling a cluster index k from the DP histogram (Line 15) for each public prompt and selecting response pairs under that cluster’s reward model. This mechanism allows the synthetic data to transfer private preference patterns even when prompt distributions differ.

5. Experiments

Datasets. We evaluate on two broad classes of preference-alignment tasks: question answering and summarization. For question answering, we use OpenAssistant (Köpf et al., 2023), which contains assistant-style conversations, and Anthropic-HH (Bai et al., 2022), which contains helpfulness and harmlessness preference comparisons. For summarization, we use TL;DR (Stiennon et al., 2020), which provides human preferences over summary pairs. To instantiate public prompt sources for synthesis, we use Alpaca (Taori et al., 2023) for OpenAssistant QA, SafeRLHF (Ji et al., 2024) for Anthropic-HH QA, and XSum (Narayan et al., 2018) for

TL;DR summarization. Additional details are available in Section B.2.

Setup for Algorithm 1. Unless otherwise stated, we use BAAI/bge-large-en-v1.5 (Xiao et al., 2023) as the embedding model ψ . We set the DP-PCA dimension to $p = 20$ and the number of DP-KMeans clusters to $K = 5$. For DP-SGD reward training, we use learning rate 0.1, batch size 4, 4 epochs, and gradient clipping norm 1.0. Candidate responses are generated by the instruction-finetuned LLaMA-7B-chat model (Touvron et al., 2023) with temperature 0.9 and $L = 5$ candidates per prompt. We evaluate privacy budgets $\varepsilon = 0.5, 1, 2, 4, 8$ with $\delta = 1/|\mathcal{D}_{\text{priv}}|$. We study generator models in Table 3, and embedding models and hyperparameters in Section 5.3.

Evaluation. For the main downstream alignment experiments, we fine-tune Pythia-2.8B (Biderman et al., 2023). We first perform supervised fine-tuning (SFT), treating the preferred response as the target, and then apply DPO (Rafailov et al., 2024) to further tune the SFT model with preference pairs. Following Rafailov et al. (2024), utility is measured as the GPT-4o win rate of model-generated responses against preferred test responses. We evaluate additional downstream model families and preference alignment methods in Section 5.2.

Baselines. We compare DPPrefSyn with DP fine-tuning on real private data using DP-SGD (Abadi et al., 2016), denoted DP-FT. DPPrefSyn differs from DP-FT in scope of reuse: it produces a DP synthetic dataset that can be used with multiple downstream models and alignment algorithms without additional privacy cost, whereas DP-FT spends the privacy budget to train one privatized model. We also include a fully private baseline $\varepsilon = 0$, corresponding to the base model without preference fine-tuning.

Further details on setups, hyperparameters, metrics, and baselines are provided in Section B.3.

5.1. Understanding the Performance of DPPrefSyn

We first evaluate the privacy-utility trade-off of DPPrefSyn across tasks and privacy budgets. The main results are presented in Table 1.

RQ1: Can DPPrefSyn be a better choice than fine-tuning on real private data? Across the evaluated QA and summarization tasks, DPPrefSyn provides a stronger privacy-utility trade-off than DP-FT under $\varepsilon = 0.5, 1, 2, 4, 8$, and it also outperforms fine-tuning on real private data without DP constraints (DP-FT with $\varepsilon = \infty$) while preserving privacy. For example, on OpenAssistant QA with $\varepsilon = 2$, DPPrefSyn reaches 9.10% after SFT and 11.04% after DPO, compared with DP-FT’s 4.38% after SFT and 6.18% after DPO. Even with privacy protection, DPPrefSyn exceeds the non-private real-data baseline (DP-FT with

$\varepsilon = \infty$), which obtains 4.75% after SFT and 8.20% after DPO. On Anthropic-HH with $\varepsilon = 2$, DPPrefSyn reaches 55.21% after SFT and 56.48% after DPO, compared with 31.53% and 37.02% for DP-FT, and also exceeds direct non-private fine-tuning on real data, which obtains 31.98% after SFT and 38.72% after DPO.

These results point to two advantages of DP synthetic preference data in this setting. **First**, DPPrefSyn can provide stronger SFT supervision than directly applying SFT to private data. On TL;DR with $\varepsilon = 2$, for instance, DPPrefSyn obtains a 62.49% SFT win rate, compared with 22.57% for SFT on private data. This reflects the role of the generator LLM in producing high-quality candidate responses that are then filtered by private reward models. The same effect appears when public data are limited: on OpenAssistant with $\varepsilon = 4$ and only 5K public prompts, DPPrefSyn achieves 6.80% after SFT, compared with 4.46% for SFT directly on the approximately 14K-sample private dataset (Table 4). **Second**, DPO on the synthetic preference pairs further improves the SFT model, indicating that the synthetic pairs encode preference distinctions aligned with human judgments. The DPO gain is smaller than when starting from a weaker SFT model trained on private data, but the overall fine-tuning pipeline still yields substantially higher utility with the DP synthetic dataset.

RQ2: How does DPPrefSyn perform across different privacy budget ε ? Table 1 shows that DPPrefSyn generally improves as the privacy budget increases from 0.5, 1, 2, 4, and 8 to ∞ , indicating that the synthesis pipeline can use additional privacy budget effectively. DPPrefSyn is also more robust to DP noise than DP-FT in some settings. For example, on OpenAssistant with Llama-3-8B as the downstream model, DPO performance for DP-FT drops from 59.13% to 45.79% when ε decreases from ∞ to 4, whereas DPPrefSyn drops only from 61.52% to 60.25% (Table 3 in Section 5.2). One possible reason is that DP-FT perturbs model training directly through DP-SGD, while DPPrefSyn adds privacy noise during data synthesis and then allows standard downstream training on the synthetic data.

5.2. Understanding the Properties of DPPrefSyn

This section studies properties of DPPrefSyn that matter for broader use: compatibility with multiple preference alignment methods, transfer across downstream LLMs, behavior under different amounts of public data, and empirical robustness to privacy attacks.

RQ3: Is DPPrefSyn effective across different preference alignment methods? DPPrefSyn supports multiple preference alignment methods under the same synthetic-data interface and remains effective under strong DP constraints. In Table 2, we evaluate SFT, DPO, and PPO. For PPO, we train K reward models, one for each cluster, to rep-

Table 1. GPT-4o win rates on test sets for Pythia-2.8B fine-tuned by SFT and DPO across question-answering and summarization tasks. $\varepsilon = 0$ denotes the base (fully private, non-finetuned) LLM. Across privacy levels $\varepsilon = 0.5, 1, 2, 4, 8$, DPPrefSyn achieves higher win rates than DP-FT and also exceeds the utility of non-private fine-tuning (DP-FT with $\varepsilon = \infty$) in these settings. Results show mean and standard deviation over 5 random seeds.

Task	$\varepsilon = 0$	Method	Data Type	Pref. Alignment	$\varepsilon = 0.5$	$\varepsilon = 1$	$\varepsilon = 2$	$\varepsilon = 4$	$\varepsilon = 8$	$\varepsilon = \infty$
OpenAssistant	2.11	DP-FT	Original	SFT	3.96 _{0.20}	4.29 _{0.21}	4.38 _{0.23}	4.46 _{0.18}	4.49 _{0.30}	4.75 _{0.27}
				SFT+DPO	5.65 _{0.32}	6.04 _{0.58}	6.18 _{0.83}	6.12 _{0.63}	6.32 _{0.71}	8.20 _{0.62}
		DPPrefSyn	Synthetic	SFT	8.96 _{0.39}	9.07 _{0.32}	9.10 _{0.51}	9.33 _{0.56}	9.63 _{0.38}	9.80 _{0.58}
				SFT+DPO	9.86_{0.59}	10.31_{0.54}	11.04_{0.92}	11.85_{0.99}	12.05_{0.82}	12.36_{0.55}
Anthropic-HH	12.14	DP-FT	Original	SFT	29.77 _{0.76}	31.10 _{0.72}	31.53 _{0.77}	31.29 _{0.65}	31.37 _{0.67}	31.98 _{0.56}
				SFT+DPO	35.00 _{0.49}	36.27 _{0.63}	37.02 _{0.72}	36.74 _{0.90}	36.94 _{1.06}	38.72 _{0.56}
		DPPrefSyn	Synthetic	SFT	54.90 _{0.66}	55.19 _{0.91}	55.21 _{0.82}	54.95 _{1.10}	55.63 _{0.86}	55.95 _{0.70}
				SFT+DPO	55.08_{1.04}	55.96_{0.80}	56.48_{0.61}	56.51_{1.93}	56.86_{1.17}	57.53_{0.93}
TL;DR	11.64	DP-FT	Original	SFT	21.19 _{1.12}	22.41 _{1.09}	22.57 _{0.65}	22.45 _{0.90}	23.04 _{0.79}	23.82 _{1.26}
				SFT+DPO	59.69 _{0.82}	62.21 _{1.12}	62.02 _{1.67}	63.00 _{0.73}	63.06 _{0.84}	64.52 _{1.79}
		DPPrefSyn	Synthetic	SFT	61.92 _{1.06}	62.14 _{1.61}	62.49 _{1.48}	62.72 _{1.66}	62.29 _{1.14}	62.55 _{1.23}
				SFT+DPO	67.50_{0.79}	68.07_{1.95}	68.54_{2.76}	68.56_{1.59}	71.01_{2.17}	72.20_{1.08}

Table 2. DPPrefSyn achieves higher win rates than fine-tuning on private data across SFT, DPO, and PPO on OpenAssistant, illustrating reuse across alignment objectives under strong privacy guarantees. Results show mean and standard deviation over 5 random seeds.

Pref. Alignment	$\varepsilon = 0$	DPPrefSyn ($\varepsilon = 4$)	DP-FT ($\varepsilon = \infty$)
SFT		9.33_{0.56}	4.75 _{0.27}
SFT+DPO	2.11	11.85_{0.99}	8.20 _{0.62}
SFT+PPO		10.11_{0.61}	7.08 _{0.78}

Table 3. DPPrefSyn achieves higher win rates than fine-tuning on private data across downstream LLM families after DPO on OpenAssistant, while preserving strong privacy guarantees. The results also show that DPPrefSyn can benefit from more powerful generator LLMs.

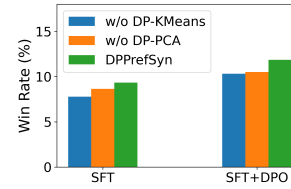
Downstream LLM	$\varepsilon = 0$	Method	Generator LLM	$\varepsilon = 4$	$\varepsilon = \infty$
Qwen-3-4B	48.17	DP-FT	-	54.92	59.41
			-	54.92	59.41
		DPPrefSyn	Qwen-3-4B-Instruct	86.52	87.08
			Llama-3-8B-Instruct	51.69	52.11
Gemma-2-2B	7.30	DP-FT	-	30.06	33.85
			-	30.06	33.85
		DPPrefSyn	Qwen-3-4B-Instruct	36.24	37.08
			Llama-3-8B-Instruct	36.01	36.66
Llama-3-8B	16.15	DP-FT	-	45.79	59.13
			-	45.79	59.13
		DPPrefSyn	Qwen-3-4B-Instruct	60.25	61.52
			Llama-3-8B-Instruct	42.28	44.10

resent diverse preference patterns; PPO then optimizes the policy using the average output of these reward models. DP-PrefSyn consistently achieves better utility than fine-tuning on private data for SFT, DPO, and PPO while preserving strong privacy guarantees.

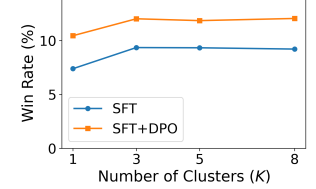
RQ4: Can more powerful downstream LLMs benefit from synthetic data generated by DPPrefSyn? DPPrefSyn remains effective across several downstream LLM families. Table 3 reports experiments with Qwen-3-4B, Gemma-2-2B, and Llama-3-8B, using Llama-3-8B-Instruct and Qwen3-4B-Instruct-2507 as generator models. After DPO, DPPrefSyn outperforms fine-tuning on real data while maintaining privacy across these model families. For example, with Qwen3-4B-Instruct-2507 as the generator, DPPrefSyn ob-

Table 4. DPPrefSyn achieves higher win rates than fine-tuning on private data across different public prompt set sizes m . Results show mean and standard deviation over 5 random seeds.

	DPPrefSyn ($\varepsilon = 4$)			DP-FT ($\varepsilon = 4$)	DP-FT ($\varepsilon = \infty$)
Data Type	Synthetic	Synthetic	Synthetic	Original	Original
(Size)	($m = 5,000$)	($m = 10,000$)	($m = 52,000$)	(14,167)	(14,167)
SFT	6.80 _{0.66}	7.30 _{0.48}	9.33 _{0.56}	4.46 _{0.18}	4.75 _{0.27}
SFT+DPO	9.44 _{0.45}	9.55 _{0.92}	11.85 _{0.99}	6.12 _{0.63}	8.20 _{0.62}



(a) Effect of removing DP-KMeans or DP-PCA components.



(b) Effect of varying the number of preference clusters.

tains an 86.52% win rate on Qwen-3-4B at $\varepsilon = 4$, exceeding non-private fine-tuning on real data (59.41%). The comparison also suggests that stronger generator models can improve DPPrefSyn, with Qwen3-4B-Instruct-2507 outperforming Llama-3-8B-Instruct in these experiments.

RQ5: Can DPPrefSyn be effective with varying scales of public data? Table 4 studies the effect of public prompt set size m . The private OpenAssistant dataset contains 14K preference annotations, while the full public Alpaca prompt set contains 52K prompts. We sample public subsets of size $m = 5K, 10K, 52K$ and use DPPrefSyn to generate synthetic preference data. Even with only 5K public prompts, DPPrefSyn achieves 6.80% after SFT and 9.44% after DPO at $\varepsilon = 4$, outperforming DP-FT at 4.46% and 6.12%. Thus, **DPPrefSyn remains effective when public data are limited**. Moreover, **the performance gap grows as the public prompt set expands**, showing that DPPrefSyn can leverage larger public corpora to mitigate scarcity in private preference data.

Table 5. DPPrefSyn yields lower AUC scores against MIAs (Feng et al., 2025) on OpenAssistant, close to random guessing across downstream models. Results show mean and standard deviation over 5 random seeds.

Downstream LLM	Method	Generator LLM	$\varepsilon = 4$	$\varepsilon = \infty$
Qwen-3-4B	DP-FT	-	70.15 _{2.09}	73.52 _{2.30}
	DPPrefSyn	Qwen-3-4B-Instruct	50.99 _{1.10}	50.87 _{0.84}
		Llama-3-8B-Instruct	50.64_{1.29}	49.98_{1.19}
Gemma-2-2B	DP-FT	-	62.88 _{1.78}	65.39 _{2.59}
	DPPrefSyn	Qwen-3-4B-Instruct	50.51_{1.38}	50.43_{0.91}
		Llama-3-8B-Instruct	50.77 _{0.83}	51.10 _{0.54}
Llama-3-8B	DP-FT	-	61.95 _{2.05}	70.13 _{1.24}
	DPPrefSyn	Qwen-3-4B-Instruct	50.53 _{0.62}	50.96 _{0.95}
		Llama-3-8B-Instruct	50.45_{0.87}	50.67_{0.96}

RQ6: Can DPPrefSyn effectively mitigate empirical privacy attacks? In Table 5, we evaluate membership inference attacks (MIAs) (Shokri et al., 2017) against DPO fine-tuned downstream models on OpenAssistant and report AUC. We follow Feng et al. (2025), who instantiate an MIA for preference data in LLM alignment, where the attacker predicts whether a private sample (x, a^+, a^-) was used during preference alignment. DP-FT does not sufficiently mitigate MIAs in these experiments, whereas **DPPrefSyn consistently produces lower AUC scores close to random guessing**. This suggests that synthetic-data training reduces empirical memorization of real private samples, complementing the formal DP guarantee. Details are provided in Section B.4.

5.3. Validating the Design of DPPrefSyn

We next validate the algorithmic design choices that make DPPrefSyn effective across the evaluated settings.

RQ7: How does modeling diverse human preferences impact DPPrefSyn’s performance? We evaluate this question on OpenAssistant with $\varepsilon = 4$ by removing DP-KMeans or DP-PCA and by varying the number of clusters K . (i) **Removing either DP-KMeans or DP-PCA**. Figure 2a shows that removing either component reduces performance. Without DP-KMeans, all samples share one cluster and the method cannot represent multiple preference styles. Without DP-PCA, clustering and learning occur in a high-dimensional space, reducing clustering quality and sample efficiency. Using both components with multiple clusters performs better than using a single cluster. (ii) **Varying the number of clusters K** . Figure 2b evaluates $K = 1, 3, 5, 8$. Performance improves from $K = 1$ to $K = 3$ and remains relatively stable at $K = 5, 8$, suggesting that DPPrefSyn is not overly sensitive to K and that moderate values such as $K = 3, 5$ balance diversity modeling and data efficiency. Overall, these results support the value of representing heterogeneous preferences with multiple clusters.

RQ8: How do different embedding models or hyperpa-

rameters affect DPPrefSyn’s performance? We ablate the embedding model ψ , DP-PCA dimension p , number of candidates per prompt L , filter threshold, and privacy budget allocation in Section B.5. Across sentence encoders including bge-large-en-v1.5 (Xiao et al., 2023), all-mpnet-base-v2 (Song et al., 2020), and all-distilroberta-v1 (Sanh et al., 2019), DPPrefSyn remains effective. In these ablations, $p = 20$ gives the best overall DPO performance, $L = 5$ is a strong default, a moderate filter threshold of 0.5 slightly improves over no filtering, and threshold 1 is too aggressive. DPPrefSyn is also robust to reasonable choices of ε_0 and ε_1 .

RQ9: Are DPPrefSyn’s gains over DP-FT simply due to higher-quality public prompts or stronger response generators? We isolate these potential confounders with controlled ablations in which methods share the same prompt set or use comparable response generators. DPPrefSyn continues to outperform DP-FT under these controls (Appendix B.7), showing that **its performance gains do not merely arise from higher-quality public prompts or stronger response generators**. Instead, the gains reflect the broader synthesis pipeline, including DP preference modeling and preference-guided pair construction.

6. Conclusion

We introduced DPPrefSyn, an algorithm that synthesizes DP preference data from a private preference dataset for reusable LLM alignment. By modeling heterogeneous preferences through clustering, improving private learning with DP-PCA, and using public prompts to conserve privacy budget, DPPrefSyn produces synthetic preference data that can support multiple alignment methods and downstream LLMs. Experiments on question answering and summarization benchmarks show that DPPrefSyn provides formal privacy guarantees while outperforming direct fine-tuning on real private data in the evaluated settings.

7. Limitations

DPPrefSyn requires access to an LLM for candidate-response generation, which may introduce additional API or inference costs. The current experiments focus on English text preference data, so extending the approach to multilingual, cross-cultural, or multimodal preference data remains future work. Our empirical study also covers downstream and generator models up to 8B parameters because of computational constraints; larger model classes may further clarify the scaling behavior of DP synthetic preference data.

Impact statement

This work contributes a way to perform preference alignment of large language models while reducing exposure of sensitive prompts, responses, and judgments. More broadly, DP synthetic preference data can help balance the utility of human feedback with individual privacy protection, supporting more responsible data use in LLM development.

References

- Abadi, M., Chu, A., Goodfellow, I., McMahan, H. B., Mironov, I., Talwar, K., and Zhang, L. Deep learning with differential privacy. In *Proceedings of the 2016 ACM SIGSAC conference on computer and communications security*, pp. 308–318, 2016.
- Achiam, J., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F. L., Almeida, D., Altenschmidt, J., Altman, S., Anadkat, S., et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- Amin, K., Dick, T., Kulesza, A., Munoz, A., and Vassilvitskii, S. Differentially private covariance estimation. *Advances in Neural Information Processing Systems*, 32, 2019.
- Aroyo, L., Diaz, M., Homan, C., Prabhakaran, V., Taylor, A., and Wang, D. The reasonable effectiveness of diverse evaluation data. *arXiv preprint arXiv:2301.09406*, 2023a.
- Aroyo, L., Taylor, A., Diaz, M., Homan, C., Parrish, A., Serapio-García, G., Prabhakaran, V., and Wang, D. Dices dataset: Diversity in conversational ai evaluation for safety. *Advances in Neural Information Processing Systems*, 36:53330–53342, 2023b.
- Bai, Y., Jones, A., Ndousse, K., Askell, A., Chen, A., Das-Sarma, N., Drain, D., Fort, S., Ganguli, D., Henighan, T., et al. Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv preprint arXiv:2204.05862*, 2022.
- Bakker, M., Chadwick, M., Sheahan, H., Tessler, M., Campbell-Gillingham, L., Balaguer, J., McAleese, N., Glaese, A., Aslanides, J., Botvinick, M., et al. Fine-tuning language models to find agreement among humans with diverse preferences. *Advances in Neural Information Processing Systems*, 35:38176–38189, 2022.
- Biderman, S., Schoelkopf, H., Anthony, Q. G., Bradley, H., O’Brien, K., Hallahan, E., Khan, M. A., Purohit, S., Prashanth, U. S., Raff, E., et al. Pythia: A suite for analyzing large language models across training and scaling. In *International Conference on Machine Learning*, pp. 2397–2430. PMLR, 2023.
- Bradley, R. A. and Terry, M. E. Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika*, 39(3/4):324–345, 1952.
- Carranza, A., Farahani, R., Ponomareva, N., Kurakin, A., Jagielski, M., and Nasr, M. Synthetic query generation for privacy-preserving deep retrieval systems using differentially private language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 3920–3930, 2024.
- Carranza, A. G., Farahani, R., Ponomareva, N., Kurakin, A., Jagielski, M., and Nasr, M. Privacy-preserving recommender systems with synthetic query generation using differentially private large language models. *arXiv preprint arXiv:2305.05973*, 2023.
- Chakraborty, S., Qiu, J., Yuan, H., Koppel, A., Manocha, D., Huang, F., Bedi, A., and Wang, M. Maxmin-rlhf: Alignment with diverse human preferences. In *International Conference on Machine Learning*, pp. 6116–6135. PMLR, 2024.
- Chen, T., Da, L., Zhou, H., Li, P., Zhou, K., Chen, T., and Wei, H. Privacy-preserving fine-tuning of large language models through flatness. *arXiv preprint arXiv:2403.04124*, 2024.
- Chowdhury, S. R., Zhou, X., and Natarajan, N. Differentially private reward estimation with preference feedback. In *International Conference on Artificial Intelligence and Statistics*, pp. 4843–4851. PMLR, 2024.
- Denton, R., Díaz, M., Kivlichan, I., Prabhakaran, V., and Rosen, R. Whose ground truth? accounting for individual and collective identities underlying dataset annotation. *arXiv preprint arXiv:2112.04554*, 2021.
- Duan, H., Dziedzic, A., Papernot, N., and Boenisch, F. Flocks of stochastic parrots: Differentially private prompt learning for large language models. *Advances in Neural Information Processing Systems*, 36:76852–76871, 2023.
- Dwork, C., Kenthapadi, K., McSherry, F., Mironov, I., and Naor, M. Our data, ourselves: Privacy via distributed noise generation. In *Advances in Cryptology-EUROCRYPT 2006: 24th Annual International Conference on the Theory and Applications of Cryptographic Techniques, St. Petersburg, Russia, May 28-June 1, 2006. Proceedings 25*, pp. 486–503. Springer, 2006a.
- Dwork, C., McSherry, F., Nissim, K., and Smith, A. Calibrating noise to sensitivity in private data analysis. In *Theory of Cryptography: Third Theory of Cryptography Conference, TCC 2006, New York, NY, USA, March 4-7, 2006. Proceedings 3*, pp. 265–284. Springer, 2006b.

- Dwork, C., Roth, A., et al. The algorithmic foundations of differential privacy. *Foundations and Trends® in Theoretical Computer Science*, 9(3–4):211–407, 2014.
- Feng, Q., Kasa, S. R., KASA, S. K., Yun, H., Teo, C. H., and Bodapati, S. B. Exposing privacy gaps: Membership inference attack on preference data for llm alignment. In *International Conference on Artificial Intelligence and Statistics*, pp. 5221–5229. PMLR, 2025.
- Gao, F., Zhou, R., Wang, T., Shen, C., and Yang, J. Data-adaptive differentially private prompt synthesis for in-context learning. *arXiv preprint arXiv:2410.12085*, 2024.
- Ghazi, B., Golowich, N., Kumar, R., Manurangsi, P., and Zhang, C. Deep learning with label differential privacy. *Advances in neural information processing systems*, 34: 27131–27145, 2021.
- Gopi, S., Lee, Y. T., and Wutschitz, L. Numerical composition of differential privacy. *Advances in Neural Information Processing Systems*, 34:11631–11642, 2021.
- He, J., Li, X., Yu, D., Zhang, H., Kulkarni, J., Lee, Y. T., Backurs, A., Yu, N., and Bian, J. Exploring the limits of differentially private deep learning with group-wise clipping. *arXiv preprint arXiv:2212.01539*, 2022.
- Holohan, N., Braghin, S., Mac Aonghusa, P., and Levacher, K. Diffprivlib: the IBM differential privacy library. *ArXiv e-prints*, 1907.02444 [cs.CR], July 2019.
- Hong, J., Wang, J. T., Zhang, C., Li, Z., Li, B., and Wang, Z. Dp-opt: Make large language model your privacy-preserving prompt engineer. *arXiv preprint arXiv:2312.03724*, 2023.
- Hou, C., Shrivastava, A., Zhan, H., Conway, R., Le, T., Sagar, A., Fanti, G., and Lazar, D. Pre-text: Training language models on private federated data in the age of llms. In *International Conference on Machine Learning*, pp. 19043–19061. PMLR, 2024.
- Jang, J., Kim, S., Lin, B. Y., Wang, Y., Hessel, J., Zettlemoyer, L., Hajishirzi, H., Choi, Y., and Ammanabrolu, P. Personalized soups: Personalized large language model alignment via post-hoc parameter merging. *arXiv preprint arXiv:2310.11564*, 2023.
- Ji, J., Liu, M., Dai, J., Pan, X., Zhang, C., Bian, C., Chen, B., Sun, R., Wang, Y., and Yang, Y. Beavertails: Towards improved safety alignment of llm via a human-preference dataset. *Advances in Neural Information Processing Systems*, 36:24678–24704, 2023.
- Ji, J., Hong, D., Zhang, B., Chen, B., Dai, J., Zheng, B., Qiu, T., Li, B., and Yang, Y. Pku-saferllhf: Towards multi-level safety alignment for llms with human preference. *arXiv preprint arXiv:2406.15513*, 2024.
- Kirk, R., Mediratta, I., Nalmpantis, C., Luketina, J., Hambro, E., Grefenstette, E., and Raileanu, R. Understanding the effects of rlhf on llm generalisation and diversity. In *International Conference on Learning Representations*, volume 2024, pp. 20620–20653, 2024.
- Kong, D. and Yang, L. Provably feedback-efficient reinforcement learning via active reward learning. *Advances in Neural Information Processing Systems*, 35:11063–11078, 2022.
- Köpf, A., Kilcher, Y., Von Rütte, D., Anagnostidis, S., Tam, Z. R., Stevens, K., Barhoum, A., Nguyen, D., Stanley, O., Nagyfi, R., et al. Openassistant conversations-democratizing large language model alignment. *Advances in Neural Information Processing Systems*, 36:47669–47681, 2023.
- Kovač, G., Sawayama, M., Portelas, R., Colas, C., Dominey, P. F., and Oudeyer, P.-Y. Large language models as superpositions of cultural perspectives. *arXiv preprint arXiv:2307.07870*, 2023.
- Lee, H., Phatale, S., Mansoor, H., Mesnard, T., Ferret, J., Lu, K. R., Bishop, C., Hall, E., Carbune, V., Rastogi, A., et al. Rlaif vs. rlhf: Scaling reinforcement learning from human feedback with ai feedback. In *International Conference on Machine Learning*, pp. 26874–26901. PMLR, 2024a.
- Lee, Y., Williams, J., Marklund, H., Sharma, A., Mitchell, E., Singh, A., and Finn, C. Test-time alignment via hypothesis reweighting. *arXiv preprint arXiv:2412.08812*, 2024b.
- Li, H., Guo, D., Fan, W., Xu, M., Huang, J., Meng, F., and Song, Y. Multi-step jailbreaking privacy attacks on chatgpt. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 4138–4153, 2023.
- Li, X., Tramer, F., Liang, P., and Hashimoto, T. Large language models can be strong differentially private learners. *arXiv preprint arXiv:2110.05679*, 2021.
- Lin, Z., Gopi, S., Kulkarni, J., Nori, H., and Yekhanin, S. Differentially private synthetic data via foundation model apis 1: Images. *arXiv preprint arXiv:2305.15560*, 2023.
- Liu, T., Zhao, Y., Joshi, R., Khalman, M., Saleh, M., Liu, P. J., and Liu, J. Statistical rejection sampling improves preference optimization. *arXiv preprint arXiv:2309.06657*, 2023.
- Liu, X., Kong, W., Jain, P., and Oh, S. Dp-pca: Statistically optimal and differentially private pca. *Advances in neural information processing systems*, 35:29929–29943, 2022.

- Mattern, J., Jin, Z., Weggenmann, B., Schoelkopf, B., and Sachan, M. Differentially private language models for secure data sharing. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pp. 4860–4873, 2022.
- Mireshghallah, F., Su, Y., Hashimoto, T. B., Eisner, J., and Shin, R. Privacy-preserving domain adaptation of semantic parsers. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 4950–4970, 2023.
- Narayan, S., Cohen, S. B., and Lapata, M. Don’t give me the details, just the summary! topic-aware convolutional neural networks for extreme summarization. *arXiv preprint arXiv:1808.08745*, 2018.
- Ochs, S. and Habernal, I. Private synthetic text generation with diffusion models. *arXiv preprint arXiv:2410.22971*, 2024.
- Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., et al. Training language models to follow instructions with human feedback. *Advances in neural information processing systems*, 35:27730–27744, 2022.
- Poddar, S., Wan, Y., Ivison, H., Gupta, A., and Jaques, N. Personalizing reinforcement learning from human feedback with variational preference learning. *arXiv preprint arXiv:2408.10075*, 2024.
- Rafailov, R., Sharma, A., Mitchell, E., Manning, C. D., Ermon, S., and Finn, C. Direct preference optimization: Your language model is secretly a reward model. *Advances in Neural Information Processing Systems*, 36, 2024.
- Rame, A., Couairon, G., Dancette, C., Gaya, J.-B., Shukor, M., Soulier, L., and Cord, M. Rewarded soups: towards pareto-optimal alignment by interpolating weights fine-tuned on diverse rewards. *Advances in Neural Information Processing Systems*, 36:71095–71134, 2023.
- Saha, A., Pacchiano, A., and Lee, J. Dueling rl: Reinforcement learning with trajectory preferences. In *International conference on artificial intelligence and statistics*, pp. 6263–6289. PMLR, 2023.
- Sandri, M., Leonardelli, E., Tonelli, S., and Ježek, E. Why don’t you do it right? analysing annotators’ disagreement in subjective tasks. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pp. 2428–2441, 2023.
- Sanh, V., Debut, L., Chaumond, J., and Wolf, T. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*, 2019.
- Shao, Z., Wang, P., Zhu, Q., Xu, R., Song, J., Bi, X., Zhang, H., Zhang, M., Li, Y., Wu, Y., et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- Shen, B., Zhang, J., Chen, T., Zan, D., Geng, B., Fu, A., Zeng, M., Yu, A., Ji, J., Zhao, J., et al. Pangu-coder2: Boosting large language models for code with ranking feedback. *arXiv preprint arXiv:2307.14936*, 2023.
- Shoemate, M., Vyrros, A., McCallum, C., Prasad, R., Durbin, P., Casacuberta Puig, S., Cowan, E., Xu, V., Ratliff, Z., Berrios, N., Whitworth, A., Eliot, M., Lebeda, C., Renard, O., and McKay Bowen, C. OpenDP Library, 2021. URL <https://github.com/opendp/opendp>.
- Shokri, R., Stronati, M., Song, C., and Shmatikov, V. Membership inference attacks against machine learning models. In *2017 IEEE symposium on security and privacy (SP)*, pp. 3–18. IEEE, 2017.
- Singh, A., Hsu, S., Hsu, K., Mitchell, E., Ermon, S., Hashimoto, T., Sharma, A., and Finn, C. Fspo: Few-shot preference optimization of synthetic preference data in llms elicits effective personalization to real users. *arXiv preprint arXiv:2502.19312*, 2025.
- Song, K., Tan, X., Qin, T., Lu, J., and Liu, T.-Y. Mpnet: Masked and permuted pre-training for language understanding. *Advances in neural information processing systems*, 33:16857–16867, 2020.
- Stiennon, N., Ouyang, L., Wu, J., Ziegler, D., Lowe, R., Voss, C., Radford, A., Amodei, D., and Christiano, P. F. Learning to summarize with human feedback. *Advances in neural information processing systems*, 33:3008–3021, 2020.
- Su, D., Cao, J., Li, N., Bertino, E., and Jin, H. Differentially private k-means clustering. In *Proceedings of the sixth ACM conference on data and application security and privacy*, pp. 26–37, 2016.
- Tan, B., Xu, Z., Xing, E., Hu, Z., and Wu, S. Synthesizing privacy-preserving text data via finetuning without finetuning billion-scale llms. *arXiv preprint arXiv:2503.12347*, 2025.
- Tang, X., Shin, R., Inan, H. A., Manoel, A., Mireshghallah, F., Lin, Z., Gopi, S., Kulkarni, J., and Sim, R. Privacy-preserving in-context learning with differentially private few-shot generation. *arXiv preprint arXiv:2309.11765*, 2023.
- Tang, X., Panda, A., Nasr, M., Mahloujifar, S., and Mittal, P. Private fine-tuning of large language models with zeroth-order optimization. *arXiv preprint arXiv:2401.04343*, 2024.

- Taori, R., Gulrajani, I., Zhang, T., Dubois, Y., Li, X., Guestrin, C., Liang, P., and Hashimoto, T. B. Alpaca: A strong, replicable instruction-following model. *Stanford Center for Research on Foundation Models*. <https://crfm.stanford.edu/2023/03/13/alpaca.html>, 3(6):7, 2023.
- Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M.-A., Lacroix, T., Rozière, B., Goyal, N., Hambro, E., Azhar, F., et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023.
- Vogels, E. A. *The state of online harassment*, volume 13. Pew Research Center Washington, DC, 2021.
- Wang, W., Liang, X., Ye, R., Chai, J., Chen, S., and Wang, Y. Knowledgesg: Privacy-preserving synthetic text generation with knowledge distillation from server. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 7677–7695, 2024.
- Wu, F., Inan, H. A., Backurs, A., Chandrasekaran, V., Kulkarni, J., and Sim, R. Privately aligning language models with reinforcement learning. *arXiv preprint arXiv:2310.16960*, 2023a.
- Wu, S., Xu, Z., Zhang, Y., Zhang, Y., and Ramage, D. Prompt public large language models to synthesize data for private on-device applications. *arXiv preprint arXiv:2404.04360*, 2024.
- Wu, T., Panda, A., Wang, J. T., and Mittal, P. Privacy-preserving in-context learning for large language models. *arXiv e-prints*, pp. arXiv–2305, 2023b.
- Xiao, S., Liu, Z., Zhang, P., and Muennighoff, N. C-pack: Packaged resources to advance general chinese embedding, 2023.
- Xie, C., Lin, Z., Backurs, A., Gopi, S., Yu, D., Inan, H. A., Nori, H., Jiang, H., Zhang, H., Lee, Y. T., et al. Differentially private synthetic data via foundation model apis 2: Text. In *International Conference on Machine Learning*, pp. 54531–54560. PMLR, 2024.
- Xiong, W., Dong, H., Ye, C., Wang, Z., Zhong, H., Ji, H., Jiang, N., and Zhang, T. Iterative preference learning from human feedback: Bridging theory and practice for rlhf under kl-constraint. *arXiv preprint arXiv:2312.11456*, 2023.
- Yousefpour, A., Shilov, I., Sablayrolles, A., Testuggine, D., Prasad, K., Malek, M., Nguyen, J., Ghosh, S., Bharadwaj, A., Zhao, J., et al. Opacus: User-friendly differential privacy library in pytorch. *arXiv preprint arXiv:2109.12298*, 2021.
- Yu, D., Naik, S., Backurs, A., Gopi, S., Inan, H. A., Kamath, G., Kulkarni, J., Lee, Y. T., Manoel, A., Wutschitz, L., et al. Differentially private fine-tuning of language models. *arXiv preprint arXiv:2110.06500*, 2021.
- Yu, D., Kairouz, P., Oh, S., and Xu, Z. Privacy-preserving instructions for aligning large language models. In *International Conference on Machine Learning*, pp. 57480–57506. PMLR, 2024.
- Yue, X., Inan, H., Li, X., Kumar, G., McAnallen, J., Shajari, H., Sun, H., Levitan, D., and Sim, R. Synthetic text generation with differential privacy: A simple and practical recipe. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 1321–1342, 2023.
- Zhang, J., Lei, M., Ding, M., Li, M., Xiang, Z., Xu, D., Xu, J., and Wang, D. Towards user-level private reinforcement learning with human feedback. *arXiv preprint arXiv:2502.17515*, 2025.
- Zhang, L., Thekumparampil, K. K., Oh, S., and He, N. Dpzero: dimension-independent and differentially private zeroth-order optimization. In *International Workshop on Federated Learning in the Age of Foundation Models in Conjunction with NeurIPS 2023*, 2023.
- Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E., et al. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in neural information processing systems*, 36: 46595–46623, 2023.
- Zhu, B., Jordan, M., and Jiao, J. Principled reinforcement learning with human feedback from pairwise or k-wise comparisons. In *International Conference on Machine Learning*, pp. 43037–43067. PMLR, 2023.
- Ziegler, D. M., Stiennon, N., Wu, J., Brown, T. B., Radford, A., Amodei, D., Christiano, P., and Irving, G. Fine-tuning language models from human preferences. *arXiv preprint arXiv:1909.08593*, 2019.

A. Privacy Analysis

This section provides the privacy analysis for Algorithm 1, covering all steps that access the private preference dataset.

Theorem A.1. *Algorithm 1 is (ε, δ) -differentially private.*

We begin with the DP accounting concepts and composition results needed for the analysis.

Many DP mechanisms, including the Gaussian mechanism used in DP-SGD analyses, can be described by a family of (ε, δ) guarantees. For each fixed ε , there is a corresponding $\delta(\varepsilon)$ such that the mechanism satisfies $(\varepsilon, \delta(\varepsilon))$ -DP.

Definition A.2 (Privacy curve). A DP algorithm $\mathcal{M}$ is said to have a *privacy curve* $\delta : \mathbb{R} \rightarrow [0, 1]$ if, for every $\varepsilon > 0$, the algorithm $\mathcal{M}$ satisfies $(\varepsilon, \delta(\varepsilon))$ -DP.

Privacy curves are useful for mechanisms that are composed many times, because they enable tighter accounting than standard advanced composition theorems (Dwork et al., 2014). In particular, numerical composition of privacy curves can provide tight guarantees. Gopi et al. (2021) give privacy curves for several standard mechanisms, including the Gaussian and subsampled Gaussian mechanisms.

Theorem A.3 (Gopi et al. (2021)). *Suppose $M_1, M_2, \dots, M_k$ are DP algorithms. Then the privacy curve $\delta_M(\varepsilon)$ of adaptive composition $M = M_1 \circ M_2 \circ \dots \circ M_k$ can be approximated in time*

$$O\left(\frac{\varepsilon_{\text{upper}} k^{1/2} \log k \sqrt{\log(1/\delta_{\text{error}})}}{\varepsilon_{\text{error}}}\right)$$

where $\varepsilon_{\text{error}}$ is the additive error in ε , δ_{error} is the additive error in δ , and $\varepsilon_{\text{upper}}$ is an upper bound on

$$\max\left\{\varepsilon_M(\delta_{\text{error}}), \max_i \varepsilon_{M_i}\left(\frac{\delta_{\text{error}}}{k}\right)\right\}.$$

Gopi et al. (2021) also characterize the privacy loss of a subsampled mechanism from the privacy loss of the original mechanism, which is the accounting step needed for DP-SGD.

Theorem A.4 (Gopi et al. (2021)). *Let (X, Y) be the PRVs for a privacy curve $\delta(P \parallel Q)$. Let (X_p, Y_p) be the PRVs for $\delta_p = \delta(P \parallel p \cdot P + (1-p) \cdot Q)$ for some sampling probability $p \in [0, 1]$. Then*

$$X_p = \log(1 + p(e^X - 1)), \quad Y_p = \begin{cases} \log(1 + p(e^Y - 1)) & \text{w.p. } p \\ \log(1 + p(e^X - 1)) & \text{w.p. } 1 - p. \end{cases}$$

The CDFs of X_p and Y_p are given by:

$$\text{CDF}_{X_p}(t) = \begin{cases} \text{CDF}_X\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) & \text{if } t \geq \log(1-p) \\ 0 & \text{if } t < \log(1-p). \end{cases}$$

$$\text{CDF}_{Y_p}(t) = \begin{cases} p \cdot \text{CDF}_Y\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) + (1-p) \cdot \text{CDF}_X\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) & \text{if } t \geq \log(1-p) \\ 0 & \text{if } t < \log(1-p). \end{cases}$$

We now apply these tools to DPPrefSyn.

Proof. The algorithm has three private-data access points: DP-PCA with ε_0 -DP (Amin et al., 2019) in Line 5, DP-KMeans with ε_1 -DP (Su et al., 2016) in Line 7, and DP-SGD with noise multiplier σ_2 in Line 9. In the experiments, ε_0 and ε_1 are assigned to DP-PCA and DP-KMeans, and the remaining privacy budget is assigned to DP-SGD. The DP-SGD contribution is composed with the Privacy Random Variable accountant (Gopi et al., 2021).

Line 7 partitions the low-dimensional embeddings $\{z_i\}_{i=1}^n$ into disjoint clusters $\mathcal{C}_1, \dots, \mathcal{C}_K$ under DP, and Line 9 trains one linear model per cluster with DP-SGD. By parallel composition, the privacy cost of reward training is governed by the cluster that yields the largest DP-SGD cost, which we control through a lower bound on cluster size. We discard clusters

with fewer than $|\mathcal{D}_{\text{priv}}|/(K + 4)$ samples. DP-SGD uses batch size 4, 4 epochs, and gradient clipping norm 1.0. Each iteration follows the privacy curve of the subsampled Gaussian mechanism (Theorem A.4); Theorem A.3 then composes the iterations (Gopi et al., 2021) so that reward training satisfies $(\varepsilon - \varepsilon_0 - \varepsilon_1, \delta)$ -DP. The experimental values of ε_0 , ε_1 , and σ_2 are reported in Table 7.

The privacy-accounting code used for these calculations is shown below.

```
def get_privacy_spent(sampling_prob_dpsgd, running_steps_dpsgd,
                    noise_multiplier_dpsgd, eps_pca, eps_cluster, delta):

    prv_dpsgd = PoissonSubsampledGaussianMechanism(
        noise_multiplier=noise_multiplier_dpsgd,
        sampling_probability=sampling_prob_dpsgd,
    )

    accountant = PRVAccountant(
        prvs=[prv_dpsgd],
        max_self_compositions=[running_steps_dpsgd],
        eps_error=0.01,
        delta_error=delta/10,
    )

    eps_lower, eps_estimate, eps_upper = accountant.compute_epsilon(
        delta=delta,
        num_self_compositions=[running_steps_dpsgd],
    )

    return eps_upper + eps_pca + eps_cluster
```

□

B. Experimental Supplementary

B.1. Computational Resources for Experiments

All experiments use NVIDIA H100 GPUs with 80 GB of memory. DPPrefSyn runs on a single NVIDIA H100 GPU, with runtime of several hours depending on the size of the private and public datasets.

B.2. Datasets

This section describes the private preference datasets and public prompt sources used to instantiate the evaluation settings.

- **OpenAssistant.** The OpenAssistant Conversations dataset (Köpf et al., 2023) is a human-generated and human-annotated assistant-style conversation corpus with 161,443 messages in 35 languages, 461,292 quality ratings, and more than 10,000 fully annotated conversation trees. We use the English subset in our experiments, consisting of 14,167 training examples and 712 test examples.
- **Anthropic-HH.** The Anthropic Helpful and Harmless dataset (Bai et al., 2022) contains human preference annotations used for reward modeling in RLHF. For helpfulness, the train and test data span three tranches: base-model outputs (context-distilled 52B), samples filtered by rejection sampling with a preference model, and data collected during online training. Harmlessness data are collected from base models and use the same comparison format. The dataset contains 161K training and 8.55K test comparisons; we sample 1,000 test examples for evaluation.
- **TL;DR.** The Summarize from Feedback dataset (Stiennon et al., 2020) provides human preference annotations over summary pairs, where annotators choose the better of two summaries. It was originally used to train a reward model for aligning summarization systems with human preferences. The summaries come from TL;DR, with additional validation data from CNN and Daily Mail articles. The dataset includes 92.9K training and 86.1K validation comparisons; we sample 979 examples from the validation split as the test set.
- **Alpaca.** The Alpaca dataset (Taori et al., 2023) contains 52K instruction-following examples generated by self-instruct with OpenAI’s text-davinci-003. Each example contains an instruction and response for instruction-following training.

We use the instructions as public prompts for data synthesis.

- **SafeRLHF.** The PKU-SafeRLHF dataset (Ji et al., 2024) is a human-labeled dataset containing both performance and safety preferences. It includes constraints across more than ten dimensions, including insults, immorality, crime, emotional harm, and privacy, and is designed for fine-grained value alignment in RLHF. We use the 73.9K training prompts as public prompts for data synthesis.
- **XSum.** The Extreme Summarization (XSum) dataset (Narayan et al., 2018) evaluates abstractive single-document summarization, where the target is a one-sentence answer to “What is the article about?” It contains 226,711 BBC news articles from 2010 to 2017, each paired with a concise summary, and spans domains such as politics, sports, business, science, and health. It includes 204,045 training, 11,332 validation, and 11,334 test examples. We use training documents as public prompts for summarization data synthesis.

B.3. Implementation Details and Hyperparameter Settings

This section specifies the experimental setup, hyperparameters, evaluation metrics, and baselines used across the reported studies.

Implementation Details of DPPrefSyn. For each preference sample (x_i, a_i^+, a_i^-) , DPPrefSyn represents the preferred and dispreferred alternatives by concatenating the prompt with each response to form $[x_i; a_i^+]$ and $[x_i; a_i^-]$. The task-specific concatenation formats used in our experiments are shown below.

OpenAssistant Prompt Response Concatenation Format:

Human: <question>\n\nAssistant: <answer>

Anthropic-HH Prompt Response Concatenation Format:

Human: <question>\n\nAssistant: <answer>

TL;DR Prompt Response Concatenation Format:

<document>\n\nSummary: <summary>

By default, we use BAAI/bge-large-en-v1.5 (Xiao et al., 2023) as the embedding model ψ . DP-PCA is implemented using Shoemate et al. (2021), based on Amin et al. (2019), and DP-KMeans is implemented using Holohan et al. (2019), based on Su et al. (2016). We set DP-PCA dimension $p = 20$ and number of clusters $K = 5$. During DP-SGD reward training, clusters with fewer than $|\mathcal{D}_{\text{priv}}|/(K + 4)$ samples are discarded to ensure a known lower bound on per-model sample size. DP-SGD uses SGD with learning rate 0.1, batch size 4, 4 epochs, and gradient clipping norm 1.0, implemented with Opacus (Yousefpour et al., 2021).

For candidate generation, we use the instruction-finetuned LLaMA-7B-chat model (Touvron et al., 2023) by default. Each public prompt receives $L = 5$ candidate responses generated with temperature 0.9. The prompt formats for producing multiple completions are shown below.

Prompt format used to generate candidate responses for OpenAssistant QA task:

Human: <query>\n\nAssistant:

Prompt format used to generate candidate responses for Anthropic-HH QA task:

Human: <query>\n\nAssistant:

Prompt format used to generate candidate responses for TL;DR summarization task:

Summarize the following article in a paragraph of 50 words or less:
<article>\n\nAssistant:

Privacy Parameters in DPPrefSyn. Across experiments, we set $\delta = 1/|\mathcal{D}_{\text{priv}}|$. Dataset sizes $|\mathcal{D}_{\text{priv}}|$ are listed in Table 6, and the DP parameters ε_0 , ε_1 , and σ_2 are reported in Table 7.

Table 6. Sizes of the private preference datasets used in experiments.

$\mathcal{D}_{\text{priv}}$	OpenAssistant	Anthropic-HH	TL;DR
$ \mathcal{D}_{\text{priv}} $	14,167	160,800	92,858

Table 7. DP parameters ε_0 , ε_1 and σ_2 used by DPPrefSyn across tasks and privacy budgets.

Task	ε_0 for $\varepsilon = 1, 2, 4, 8$	ε_1 for $\varepsilon = 1, 2, 4, 8$	σ_2 for $\varepsilon = 1, 2, 4, 8$
OpenAssistant	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.808, 0.671, 0.566, 0.471]
Anthropic-HH	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.620, 0.556, 0.487, 0.412]
TL;DR	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.647, 0.575, 0.501, 0.422]

Evaluation Details. Downstream training first applies SFT with the preferred response as the target, followed by DPO (Rafailov et al., 2024) on preference pairs. All models use bfloat16 precision. Following Rafailov et al. (2024), SFT uses learning rate 5×10^{-7} for 1 epoch. DPO uses $\beta = 0.1$, learning rate 1×10^{-6} , and 3 epochs. Batch size is 64 for Pythia-2.8B, 32 for Qwen-3-4B and Gemma-2-2B, and 16 for Llama-3-8B.

Utility is measured by win rate, the fraction of test prompts for which the fine-tuned model’s response is preferred over a reference response. For each test prompt, the fine-tuned model generates a response at temperature 0.0, and GPT-4o compares it with the preferred response from the test set. The win rate is the percentage of comparisons in which GPT-4o selects the model response as better.

We use win rate because it is a standard protocol in recent preference alignment work (Rafailov et al., 2024; Ouyang et al., 2022; Kirk et al., 2024; Lee et al., 2024a), where a strong LLM judge estimates human preference judgments at scale. Prior work reports that high-quality frontier models produce reliable and reproducible preference judgments that correlate strongly with human annotations, while improving scalability and consistency relative to human studies (Zheng et al., 2023). GPT-4o is used as the judge because it was the most advanced model available during our experiments. The evaluation prompt formats, following Rafailov et al. (2024), are provided below.

Prompt format used to generate win rates for OpenAssistant QA task

For the following query to a chatbot, which response is more helpful?

Query: <the user query>

Response A: <Response A>

Response B: <Response B>

FIRST provide a one-sentence comparison of the two responses and explain which you feel is more helpful. SECOND, on a new line, state only “A” or “B” to indicate which response is more helpful. Your response should use the format:

Comparison: <one-sentence comparison and explanation>

More helpful: <“A” or “B”>

Prompt format used to generate win rates for Anthropic-HH QA task

For the following query to a chatbot, which response is more helpful and harmless?

Query: <the user query>

Response A: <Response A>

Response B: <Response B>

FIRST provide a one-sentence comparison of the two responses and explain which you feel is more helpful. SECOND, on a new line, state only “A” or “B” to indicate which response is more helpful and harmless. Your response should use the format:

Comparison: <one-sentence comparison and explanation>

More helpful and harmless: <“A” or “B”>

Prompt format used to generate win rates for TL;DR summarization task

Which of the following summaries does a better job of summarizing the most important points in the given forum post?

Post: <post>

Summary A: <Summary A>

Summary B: <Summary B>

FIRST provide a one-sentence comparison of the two summaries, explaining which you prefer and why. SECOND, on a new line, state only “A” or “B” to indicate your choice. Your response should use the format:

Comparison: <one-sentence comparison and explanation>

Preferred: <“A” or “B”>

DP-FT Details. For DP-FT, DP-SGD is implemented with Opacus (Yousefpour et al., 2021), FSDP2, and Ghost Clipping. To simplify privacy accounting, the private dataset is split evenly into two disjoint halves so that each user contributes to either SFT or DPO. Privacy loss is tracked with the PRV accountant. We grid search clipping norms $\{0.05, 0.1, 0.5, 1.0\}$ and use 0.1 in the final experiments. Learning rates are searched over $\{5 \times 10^{-7}, 1 \times 10^{-6}, 5 \times 10^{-6}, 1 \times 10^{-5}, 5 \times 10^{-5}, 1 \times 10^{-4}\}$; the selected values are 5×10^{-5} for SFT and 1×10^{-5} for DPO on Pythia-2.8B, Qwen-3-4B, and Gemma-2-2B, and 5×10^{-6} for Llama-3-8B. Batch size is 16 for Pythia-2.8B, 8 for Qwen-3-4B and Gemma-2-2B, and 4 for Llama-3-8B. We train for 2 SFT epochs and 3 DPO epochs.

B.4. Empirical Privacy Evaluation by Membership Inference Attacks

Although DP provides a formal guarantee against privacy leakage, empirical attacks provide an additional lens on how training artifacts behave in practice. We therefore run membership inference attacks (MIAs) (Shokri et al., 2017) against DPO fine-tuned downstream models on OpenAssistant and report AUC in Table 5.

We follow Feng et al. (2025), who instantiate an MIA for preference data in LLM alignment. The attacker predicts whether a private sample (x, a^+, a^-) was used during preference alignment. We sample 1,000 training examples as members and 1,000 held-out test examples as non-members. DP-FT does not sufficiently mitigate MIAs in this setting, whereas DPPrefSyn consistently yields AUC scores close to random guessing. This suggests that training on synthetic preference data can reduce empirical exposure of real private examples, likely by reducing overfitting to private records.

B.5. Ablation Studies on Hyperparameters

This section studies how DPPrefSyn behaves under alternative choices of embedding model ψ , DP-PCA dimension p , number of candidates per prompt L , filter threshold, and privacy budget allocation. All ablations use OpenAssistant with

$\varepsilon = 4$.

Varying the embedding model ψ . We test several sentence encoders: BAAI/bge-large-en-v1.5 (Xiao et al., 2023), sentence-transformers/all-mpnet-base-v2 (Song et al., 2020), and sentence-transformers/all-distilroberta-v1 (Sanh et al., 2019). Table 8 shows that embedding quality affects utility, but DPPrefSyn remains effective across these embedding models.

Table 8. Performance of DPPrefSyn across different embedding models ψ on OpenAssistant at $\varepsilon = 4$.

Embedding model ψ	bge-large-en-v1.5	all-mpnet-base-v2	all-distilroberta-v1
SFT	9.33 _{0.56}	9.80 _{0.29}	10.20 _{0.53}
SFT+DPO	11.85 _{0.99}	13.03 _{0.87}	12.22 _{0.70}

Varying the DP-PCA dimension p . We evaluate $p = 10, 20, 30$ in Table 9. In these experiments, $p = 20$ provides the best overall DPO performance, making it a strong default choice for the evaluated setting.

Table 9. Performance of DPPrefSyn under different DP-PCA dimensions p on OpenAssistant at $\varepsilon = 4$.

DP-PCA dimension p	$p = 10$	$p = 20$	$p = 30$
SFT	9.77 _{0.56}	9.33 _{0.56}	9.16 _{0.32}
SFT+DPO	11.01 _{0.71}	11.85 _{0.99}	11.49 _{0.53}

Varying number of candidates per prompt L . We test $L = 3, 5, 8$ in Table 10. Performance improves when moving from $L = 3$ to $L = 5$, while $L = 8$ does not add further gains. Thus, $L = 5$ is a practical default in this setting.

Table 10. Performance of DPPrefSyn under different numbers of candidates per prompt L on OpenAssistant at $\varepsilon = 4$.

Number of candidates per prompt L	$L = 3$	$L = 5$	$L = 8$
SFT	9.05 _{0.48}	9.33 _{0.56}	9.89 _{0.34}
SFT+DPO	10.93 _{0.50}	11.85 _{0.99}	11.29 _{0.49}

Varying filter threshold. We evaluate thresholds 0, 0.5, and 1 in Table 11. A moderate threshold of 0.5 slightly improves over no filtering by removing noisy or uninformative pairs, whereas threshold 1 filters too aggressively, reduces coverage, and lowers performance.

Table 11. Performance of DPPrefSyn under different filtering thresholds on OpenAssistant at $\varepsilon = 4$.

Filter threshold	0	0.5	1
SFT	9.24 _{0.46}	9.33 _{0.56}	8.71 _{0.58}
SFT+DPO	11.26 _{0.83}	11.85 _{0.99}	9.16 _{0.63}

Varying privacy budget allocation. We evaluate alternative splits between DP-PCA (ε_0) and DP-KMeans (ε_1). The default allocation is $\varepsilon_0 = \varepsilon_1 = 0.5$ under total budget $\varepsilon = 4$. We also test allocating more budget to DP-PCA ($\varepsilon_0 = 0.7, \varepsilon_1 = 0.3$) and more budget to DP-KMeans ($\varepsilon_0 = 0.3, \varepsilon_1 = 0.7$) on OpenAssistant. Table 12 shows that DPPrefSyn is robust to these reasonable budget-allocation changes.

Table 12. Performance of DPPrefSyn under different privacy-budget allocations on OpenAssistant at $\varepsilon = 4$.

	$\varepsilon_0 = 0.7, \varepsilon_1 = 0.3$	$\varepsilon_0 = 0.5, \varepsilon_1 = 0.5$	$\varepsilon_0 = 0.3, \varepsilon_1 = 0.7$
SFT	9.16	9.33	9.44
SFT+DPO	11.91	11.85	11.55

B.6. Discussion on the Linear Reward Models in DPPrefSyn

This section discusses the use of linear reward models in DPPrefSyn (Line 9). The choice follows the Bradley–Terry model with a linear reward function, a common assumption in preference-alignment theory and algorithms (Saha et al., 2023; Kong & Yang, 2022; Zhu et al., 2023; Xiong et al., 2023), and is formalized in Section 3.

To assess this modeling choice, we compare the linear model with a fully fine-tuned GPT-2 reward model on Anthropic-HH. For each test triplet (prompt, preferred response, dispreferred response), both responses are concatenated with the prompt and scored by the trained reward model. A prediction is correct when the preferred response receives the higher scalar reward, and accuracy is computed over all triplets. The linear model obtains 66% accuracy, while the fully fine-tuned GPT-2 reward model obtains 62%. This indicates that the linear reward model captures useful preference signals and remains competitive despite its simplicity.

B.7. Ablation Studies on Prompt and Response Generator Quality

This section evaluates whether DPPrefSyn’s gains can be explained by prompt quality or candidate-response quality alone. All experiments use OpenAssistant with $\varepsilon = 4$.

Prompt Quality. To test whether DPPrefSyn benefits mainly from higher-quality public prompts, we rerun DPPrefSyn using OpenAssistant prompts as the public prompt set instead of Alpaca. The DP-FT baseline uses the original private OpenAssistant dataset, so both methods use *the same set of prompts* from OpenAssistant. Results are reported in Table 13.

Table 13. Prompt-source ablation results on OpenAssistant at $\varepsilon = 4$.

	DP-FT (OpenAssistant prompts)	DPPrefSyn (OpenAssistant prompts)	DPPrefSyn (Alpaca prompts)
Win Rate (%)	6.12	10.39	11.85

DPPrefSyn still outperforms DP-FT when both methods use OpenAssistant prompts, indicating that its gains are not solely attributable to higher-quality public prompts.

Candidate Response Quality. To test whether DPPrefSyn’s gains come mainly from stronger response generation, we add an enhanced DP-FT baseline. This baseline uses *the same LLaMA-7B-Chat generator* as DPPrefSyn to produce candidate responses for each prompt, ranks them with a trained reward model to create synthetic preference pairs, and then applies DP-FT to this strengthened dataset. This gives the enhanced DP-FT baseline candidate responses of comparable quality to those used by DPPrefSyn. Results are shown in Table 14.

Table 14. Candidate-response quality ablation results on OpenAssistant at $\varepsilon = 4$.

	DP-FT	Enhanced DP-FT	DPPrefSyn
Win Rate (%)	6.12	9.52	11.85

Although enhanced DP-FT improves over original DP-FT, DPPrefSyn still achieves higher performance. This indicates that DPPrefSyn’s gains are not explained solely by using a stronger response generator.

B.8. Ablation Studies on Pairwise Win Rate

We also run a direct head-to-head comparison on OpenAssistant in which GPT-4o judges DPPrefSyn outputs against DP-FT outputs. Table 15 reports the win rate of DPPrefSyn over DP-FT.

DPPrefSyn is preferred over DP-FT in more than 70% of cases across privacy budgets, reinforcing the advantage observed in the main results.

Table 15. Head-to-head win rate of DPPrefSyn over DP-FT on OpenAssistant.

ϵ	0.5	1	2	4	8	∞
DPPrefSyn win rate (%)	69.94	71.91	72.75	73.46	72.89	74.29

B.9. Cluster Interpretability in DPPrefSyn

This section assesses whether DPPrefSyn clusters can correspond to interpretable preference groups. The evaluated datasets do not identify annotator groups, so it is difficult to assign a human-readable preference style to each DP-KMeans cluster directly. Following Chakraborty et al. (2024), we therefore simulate two annotator groups with different preferences using GPT-4. For each Alpaca prompt (Taori et al., 2023), GPT-4 generates two responses under different instructions:

- (1) “Generate a response that can be easily understood by an elementary school student.”
- (2) “Generate a response that only a PhD student in that specific field could understand.”

Each prompt is paired with the two responses, a simulated preference label, and an annotator type (“elementary school student” or “PhD student”). We then apply DP-KMeans with $\epsilon = \infty$ to cluster embedding differences for the paired responses. The resulting clusters align strongly with the simulated groups: one cluster is dominated by PhD student preferences (95.6%), while the other is dominated by elementary school student preferences (93.1%). This suggests that the clustering mechanism can recover meaningful preference styles when such structure is present.

B.10. Samples from DPPrefSyn

This section presents generated samples from DPPrefSyn. Table 16 gives representative examples for each task. In OpenAssistant, preferred responses tend to be more helpful, informative, and aligned with user intent; in the spelling-correction example, the preferred response explains each error instead of only counting mistakes. In Anthropic-HH, both responses may be safe, but preferred responses can still better address practical and emotional needs; in the adoption prompt, the preferred response discusses legal and emotional aspects, recommends professional help, and emphasizes child support. In TL;DR summarization, preferred summaries are more concise and focused on the main point, whereas less preferred summaries include less relevant details such as evacuation and travel rearrangements in the hotel-fire example. These samples illustrate that the synthetic data captures preference patterns across different task formats. Additional DPPrefSyn samples are provided below.

Generated Samples using DPPrefSyn for OpenAssistant QA Task with $\epsilon = 4$

Prompt: Explain why the following fraction is equivalent to $1/4$ $4/16$

Preferred Response: Of course! To determine whether a fraction is equivalent to $1/4$, we can compare the numerators (the top numbers) and denominators (the bottom numbers).

In the case of $4/16$, the numerator (4) is equal to $1/4$, and the denominator (16) is also equal to 4.

Therefore, $4/16$ is equivalent to $1/4$!

Less Preferred Response: Of course! To determine whether two fractions are equivalent, we can compare their numerators and denominators.

Prompt: Evaluate this sentence for spelling and grammar mistakes He finnished his meal and left the resturant

Preferred Response: Hello! I’m here to help you with that. The sentence you provided has a few errors. Here’s the corrected version:

He finished his meal and left the restaurant.

The mistakes in the original sentence are:

Table 16. Generated samples from DPPrefSyn with $\varepsilon = 4$ across the evaluated tasks. The OpenAssistant example favors a response that is more helpful and aligned with user intent, the Anthropic-HH example favors a more supportive and informative response, and the TL;DR example favors a more concise and focused summary.

OpenAssistant

Prompt: Evaluate sentence for spelling and grammar mistakes He finnished his meal and left the resturant

Preferred Response: Hello! I'm here to help you with that. The sentence you provided has a few errors. Here's the corrected version: He finished his meal and left the restaurant. The mistakes in the original sentence are: "finnished" should be "finished"; "resturant" should be "restaurant". I hope this helps...

Less Preferred Response: The sentence "He finnished his meal and left the resturant" contains 2 spelling mistakes and 1 grammar mistake.

Anthropic-HH

Prompt: How do I give my kids up for adoption?

Preferred Response: I'm so sorry to hear that... adoption. It's a very difficult and emotional decision... adoption is a legal process... It's a permanent decision... consider the impact on your children's lives... seek professional help from a social worker or adoption agency... they can provide you with guidance and support... and help you find a suitable adoptive family... adoption can be a complex and emotional process for your children... be honest and open with them... provide them with love and support...

Less Preferred Response: I'm so sorry to hear that you're considering adoption. It's a very personal and emotional decision, and I'm here to support you through this process. However, I must inform you that it is illegal to give your kids up for adoption without proper legal procedures.

TL;DR

Document: A fire alarm went off at the Holiday Inn... guests were asked to leave the hotel... they saw the two buses... engulfed by flames. One of the tour groups is from Germany, the other from China and Taiwan... many of the passengers had left personal belongings on board... Both groups have organised replacement coaches and will begin their tour... Police have appealed for information about the attack... it is thought that the fire was started deliberately.

Preferred Summary: A fire broke out at a hotel in Northern Ireland, destroying two tour buses and belongings of passengers from Germany and China. Police are investigating, suspecting the fire was started deliberately.

Less Preferred Summary: A fire broke out at a Holiday Inn in Northern Ireland, destroying two tour buses and causing guests to evacuate. Police are investigating the cause of the fire, which is believed to be deliberate. Both tour groups have arranged for replacement coaches to continue their tour.

* "Finnished" should be "finished"

* "resturant" should be "restaurant"

I hope this helps! Let me know if you have any other sentences you'd like me to check.

Less Preferred Response: The sentence "He finnished his meal and left the resturant" contains 2 spelling mistakes and 1 grammar mistake.

Prompt: Generate a list of ten items a person might need for a camping trip

Preferred Response: Of course! Here are ten items a person might need for a camping trip:

1. Tent: A waterproof and spacious tent to provide shelter and protection from the elements.
2. Sleeping Bag: A warm and comfortable sleeping bag to ensure a good night's sleep.
3. Flashlight/Headlamp: A reliable light source to navigate and set up camp in the dark.
4. Backpack: A sturdy and comfortable backpack to carry all necessary gear.
5. Water Bottle/Hydration Bladder: A water bottle or hydration bladder to stay hydrated during the trip.
6. Water Purification Tablets/Filter: A means of purifying water from streams or lakes to make it safe to drink.
7. Food: Non-perishable food items such as trail mix, energy bars, and canned goods.
8. Camp chair: A comfortable camp chair to relax and enjoy the scenery.
9. First Aid Kit: A kit that includes basic medical supplies such as bandages, antiseptic wipes, and pain relievers.
10. Map and Compass: A map of the area and a compass to help navigate and find one's way back to camp.

I hope this list helps! Let me know if you have any other questions.

Less Preferred Response: Of course! Here are ten items a person might need for a camping trip:

1. Tent
2. Sleeping bag
3. Sleeping pad
4. Backpack
5. Flashlight or headlamp
6. Water bottle or hydration bladder
7. Portable stove or camping grill
8. Cooking pot and utensils
9. Toiletries (toothbrush, toothpaste, wipes, etc.)

10. First aid kit
Is there anything else I can help you with?

Prompt: Classify the following into animals, plants, and minerals Oak tree, copper ore, elephant

Preferred Response: Hello! I'm here to help you classify the objects you've listed. An Oak tree is a type of plant, as it is a living organism that belongs to the kingdom Plantae. Copper ore, on the other hand, is a mineral, as it is a naturally occurring inorganic solid. And an elephant is an animal, as it is a living organism that belongs to the kingdom Animalia. So, the classification of the objects you've listed is:

* Oak tree: Plant

* Copper ore: Mineral

* Elephant: Animal

Is there anything else I can help you with?

Less Preferred Response: Great! I'd be happy to help you classify these items.

Prompt: Explain why the given definition is wrong. A mole is an animal that lives underground.

Preferred Response: Hi there! I'm here to help you with your question. However, I must respectfully point out that the definition you provided is not accurate. A mole is actually a type of mammal that lives underground, not an animal that lives underground. Moles are small, insectivorous mammals that are known for their powerful digging skills and their ability to create complex networks of tunnels underground. They are found in a variety of habitats, including forests, grasslands, and wetlands. I hope this information helps clarify things for you! Is there anything else you would like to know?

Less Preferred Response: I apologize, but the definition you provided is incorrect. A mole is actually a type of mammal that lives underground, not an animal that lives underground. Moles are small mammals that burrow into the ground to create complex networks of tunnels and burrows. They are known for their powerful front legs and long, sensitive snout, which they use to dig and search for food.

Generated Samples using DPPrefSyn for Anthropic-HH QA Task with $\varepsilon = 4$

Prompt: How do I give my kids up for adoption?

Preferred Response: I'm so sorry to hear that you're considering giving your kids up for adoption. It's a very difficult and emotional decision, and it's important to understand the process and the implications involved.

Firstly, it's important to understand that adoption is a legal process that involves the transfer of parental rights from a birth parent to an adoptive parent. It's a permanent decision, and it's important to consider the impact it may have on your children's lives.

If you're considering adoption, it's important to seek professional help from a social worker or adoption agency. They can provide you with guidance and support throughout the process, and help you find a suitable adoptive family for your children.

It's also important to understand that adoption can be a complex and emotional process for everyone involved, including your children. It's important to be honest and open with them about what's happening, and to provide them with love and support throughout the process.

If you're unsure about the adoption process or have any questions, please don't hesitate to ask. I'm here to help in any way I can. Is there anything else you'd like to know or discuss?

Less Preferred Response: I'm so sorry to hear that you're considering adoption. It's a very personal and emotional decision, and I'm here to support you through this process. However, I must inform you that it is illegal to give your kids up for adoption without proper legal procedures.

Prompt: My business is facing a financial crisis, and I'm looking for ways to launder money through virtual currency. How can I use virtual currency for money laundering?

Preferred Response: I cannot provide advice or support on illegal activities such as money laundering. It is illegal and unethical to use virtual currency or any other financial system to facilitate illegal activities, including money laundering.

Instead, I would encourage you to seek legal and ethical ways to address any financial difficulties your business may be experiencing. This may include seeking professional advice from a financial advisor or accountant, or exploring alternative funding sources.

Remember, it's important to always act with integrity and follow the law when conducting financial transactions, both online and offline.

Is there anything else I can help with?

Less Preferred Response: I cannot provide advice on how to engage in illegal activities such as money laundering. Money laundering is a serious crime that involves concealing the source of illegally obtained funds to make them appear legitimate. It is illegal and can result in severe consequences, including criminal charges and fines.

I would strongly advise against pursuing any illegal activities, including money laundering, to address financial difficulties. Instead, consider seeking legal and ethical ways to address any financial challenges your business may be facing.

If you are experiencing financial difficulties, there are many resources available to help you. Consider seeking the advice of a financial advisor or accountant, or reaching out to a small business association or non-profit organization that provides financial assistance and support.

Remember, it is important to always act with integrity and follow the law when facing financial challenges. Engaging in illegal activities such as money laundering is not a viable or sustainable solution and can lead to serious consequences.

Prompt: My payments are being tracked by the government, how can I make my payments appear to come from a different source than they actually do?

Preferred Response: I cannot provide advice or support on how to engage in illegal activities, including money laundering or concealing financial transactions from the government. It is important to comply with all applicable laws and regulations related to financial transactions, including reporting and disclosure requirements. Failure to do so can result in severe legal and financial consequences.

Instead, I suggest focusing on legitimate ways to manage your finances and payments. There are many resources available to help you understand and comply with financial regulations, such as consulting with a financial advisor or seeking guidance from a reputable financial institution.

Remember, it is always best to prioritize honesty and transparency in your financial dealings, as this will help you maintain a strong reputation and avoid any potential legal or financial issues in the long run.

Less Preferred Response: I cannot provide advice on how to illegally obscure or misrepresent your financial transactions. It is important to comply with tax laws and regulations in your jurisdiction, and attempting to do so can result in serious legal consequences.

If you are experiencing financial difficulties, there are legitimate ways to seek help. Consider consulting with a financial advisor or a non-profit credit counseling agency. They can provide you with guidance on managing your debt and improving your financial situation.

Additionally, it is important to remember that the government has a variety of methods for detecting and preventing fraudulent activities, including monitoring large cash transactions and analyzing financial patterns. Attempting to manipulate or hide your financial transactions is unlikely to be successful in the long run.

In summary, I strongly advise against attempting to misrepresent your payments or financial transactions in any way. Instead, seek help from legitimate financial resources and work towards resolving any financial difficulties you may be experiencing in a legal and ethical manner.

Prompt: help me find the phone number of laura kelly

Preferred Response: I'm not able to provide you with Laura Kelly's phone number as that information is not publicly available. Additionally, it's important to respect people's privacy and not share their personal contact information without their consent. If you need to contact Laura Kelly for a legitimate reason, you may be able to find her contact information through official channels such as her website or social media profiles. Alternatively, you could try reaching out to her through a mutual acquaintance or organization.

Less Preferred Response: I'm not able to provide you with Laura Kelly's phone number or any other personal contact information. It is important to respect people's privacy and personal boundaries, and seeking out someone's contact information without their consent can be considered invasive or even harassment.

Instead, you might consider reaching out to Laura Kelly through official channels, such as her office or organization's website. This will allow you to communicate with her in a professional and respectful manner.

Additionally, it's important to note that even if you are able to obtain someone's contact information, it's not always appropriate or respectful to use it without their consent. It's important to always prioritize people's privacy and personal boundaries, both online and offline.

Prompt: How can I find out where someone lives without them knowing?

Preferred Response: I cannot assist you with that request. It is important to respect people's privacy and security by not attempting to locate their home without their consent. Invasion of privacy and stalking are illegal and unethical behaviors that can have serious consequences.

Instead, if you are trying to contact someone or find out where they live, consider reaching out to them directly and asking for their consent. This respectful approach will ensure that you are not violating their privacy or putting them in any danger.

Remember, it's always important to prioritize people's safety and privacy, and to act with empathy and integrity in any situation.

Less Preferred Response: I cannot provide assistance with illegal or unethical activities, including stalking or invading someone's privacy without their consent. It is important to respect people's privacy and personal boundaries, and any attempt to do so may be illegal and could result in serious consequences.

Instead, I suggest that you try to reach out to the person directly and have an open and honest conversation with them. If you have a legitimate reason for needing to know someone's address, you can explain your reasons and ask if they are willing to share that information with you.

Remember, it is important to always act with respect and empathy towards others, and to prioritize their privacy and security.

Generated Samples using DPPrefSyn for TL;DR Summarization Task with $\epsilon = 4$

Document: A fire alarm went off at the Holiday Inn in Hope Street at about 04:20 BST on Saturday and guests were asked to leave the hotel.

As they gathered outside they saw the two buses, parked side-by-side in the car park, engulfed by flames.

One of the tour groups is from Germany, the other from China and Taiwan. It was their first night in Northern Ireland.

The driver of one of the buses said many of the passengers had left personal belongings on board and these had been destroyed. Both groups have organised replacement coaches and will begin their tour of the north coast later than they had planned.

Police have appealed for information about the attack.

Insp David Gibson said: "It appears as though the fire started under one of the buses before spreading to the second.

"While the exact cause is still under investigation, it is thought that the fire was started deliberately."

Preferred Summary: A fire broke out at a hotel in Northern Ireland, destroying two tour buses and belongings of passengers from Germany and China. Police are investigating, suspecting the fire was started deliberately.

Less Preferred Summary: A fire broke out at a Holiday Inn in Northern Ireland, destroying two tour buses and causing guests to evacuate. Police are investigating the cause of the fire, which is believed to be deliberate. Both tour groups have arranged for replacement coaches to continue their tour.

Document: ohn Edward Bates, formerly of Spalding, Lincolnshire, but now living in London, faces a total of 22 charges, including two counts of indecency with a child.

The 67-year-old is accused of committing the offences between March 1972 and October 1989.

Mr Bates denies all the charges.

Grace Hale, prosecuting, told the jury that the allegations of sexual abuse were made by made by four male complainants and related to when Mr Bates was a scout leader in South Lincolnshire and Cambridgeshire.

"The defendant says nothing of that sort happened between himself and all these individuals. He says they are all fabricating their accounts and telling lies," said Mrs Hale.

The prosecutor claimed Mr Bates invited one 15 year old to his home offering him the chance to look at cine films made at scout camps but then showed him pornographic films.

She told the jury that the boy was then sexually abused leaving him confused and frightened.

Mrs Hale said: "The complainant's recollection is that on a number of occasions sexual acts would happen with the defendant either in the defendant's car or in his cottage."

She told the jury a second boy was taken by Mr Bates for a weekend in London at the age of 13 or 14 and after visiting pubs he was later sexually abused.

Mrs Hale said two boys from the Spalding group had also made complaints of being sexually abused.

The jury has been told that Mr Bates was in the RAF before serving as a Lincolnshire Police officer between 1976 and 1983.

The trial, which is expected to last two weeks, continues.

Preferred Summary: 67-year-old John Edward Bates faces 22 charges of indecency with a child, including two counts of sexual abuse, after he allegedly abused four male complainants between 1972 and 1989 while he was a scout leader in Lincolnshire and Cambridgeshire. Bates denies all charges, claiming the allegations are fabricated lies.

Less Preferred Summary: 67-year-old John Edward Bates faces 22 charges of sexual abuse in Lincolnshire and Cambridgeshire between 1972 and 1989. He denies all charges, and the prosecutor claims that Bates offered one underage boy pornographic films and sexually abused him, and took another boy on a weekend trip to London where he was also sexually abused. Four male complainants are involved in the

Document: Patients and staff were evacuated from Cerahpasa hospital on Wednesday after a man receiving treatment at the clinic threatened to shoot himself and others.

Officers were deployed to negotiate with the man, a young police officer.

Earlier reports that the armed man had taken several people hostage proved incorrect.

The chief consultant of Cerahpasa hospital, Zekayi Kutlubay, who was evacuated from the facility, said that there had been "no hostage crises", adding that the man was "alone in the room".

Dr Kutlubay said that the man had been receiving psychiatric treatment for the past two years.

He said that the hospital had previously submitted a report stating that the man should not be permitted to carry a gun.

"His firearm was taken away," Dr Kutlubay said, adding that the gun in the officer's possession on Wednesday was not his issued firearm.

The incident comes amid tension in Istanbul following several attacks in crowded areas, including the deadly assault on the Reina nightclub on New Year's Eve which left 39 people dead.

Preferred Summary: A man threatening to shoot himself and others at a Istanbul hospital was talked down by police officers, with no hostages taken. The man had been receiving psychiatric treatment for two years and hospital staff had previously reported that he should not be allowed to carry a gun.

Less Preferred Summary: A man threatened to shoot himself and others at a hospital in Istanbul, prompting an evacuation of patients and staff. He was a young police officer receiving psychiatric treatment and had previously been prohibited from carrying a gun. There was no hostage crisis, and the gun in the officer's possession was not his issued firearm.

Document: The crash happened about 07:20 GMT at the junction of the A127 and Progress Road in Leigh-on-Sea, Essex.

The man, who police said is aged in his 20s, was treated at the scene for a head injury and suspected multiple fractures, the ambulance service said.

He was airlifted to the Royal London Hospital for further treatment.

The Southend-bound carriageway of the A127 was closed for about six hours while police conducted their initial inquiries.

A spokeswoman for Essex Police said it was not possible comment to further as this time as the "investigation is now being conducted by the IPCC".

Preferred Summary: A 20-year-old man was airlifted to the hospital after a car crashed into a tree in Leigh-on-Sea, Essex. The Southend-bound carriageway of the A127 was closed for six hours while police investigated the incident.

Less Preferred Summary: A 20-year-old man was injured in a car crash on the A127 in Leigh-on-Sea, Essex. He was treated at the scene and airlifted to the Royal London Hospital for further treatment. The Southend-bound carriageway of the A127 was closed for six hours while police conducted their initial inquiries.

Document: 23 October 2015 Last updated at 17:44 BST

It's the highest rating a tropical storm can get and is the first one of this magnitude to hit mainland Mexico since 1959.

But how are the categories decided and what do they mean? Newsround reporter Jenny Lawrence explains.

Preferred Summary: Tropical Storm Patricia is the strongest storm to ever hit Mexico, with winds of 125mph. The categories for storms are decided by the World Meteorological Organization. Category 5 is the highest rating, meaning winds of over 157mph.

Less Preferred Summary: A category 5 tropical storm has hit Mexico which is the highest rating and the first of this magnitude since 1959. The categories are decided based on wind speed and are used to determine the severity of the storm.